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INTEGRATED CIRCUIT AND METHOD OF MANUFACTURING SAME

Abstract

An integrated circuit includes a first and second active region, a first conductor, a first and second contact, and a set of gates. The first and second active region extend in a first direction, and is located on a first level. The first conductor extends in the first direction and a second direction, is located on a lower portion of the first level, and is between the first and second active region. The first contact extends in the second direction, overlaps the first conductor and is located on a second level. The second contact extends in the second direction, overlaps the first conductor, is located on the second level, and is separated from the first contact in the first direction. The first conductor electrically couples the first contact and the second contact together.

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Background/Summary

PRIORITY CLAIM [0001] The present application is a continuation of U.S. application Ser. No. 18/053,012, filed Nov. 7, 2022, which is a continuation of U.S. application Ser. No. 16/936,249, filed Jul. 22, 2020, now U.S. Pat. No. 11,508,661, issued Nov. 22, 2022, which is a divisional of U.S. application Ser. No. 16/135,684, filed Sep. 19, 2018, now U.S. Pat. No. 10,734,321, issued Aug. 4, 2020, which claims the benefit of U.S. Provisional Application No. 62/564,663, filed Sep. 28, 2017, which are herein incorporated by reference in their entireties.

BACKGROUND

[0002] The recent trend in miniaturizing integrated circuits (ICs) has resulted in smaller devices which consume less power yet provide more functionality at higher speeds. The miniaturization process has also resulted in stricter design and manufacturing specifications as well as reliability challenges. Various electronic design automation (EDA) tools generate, optimize and verify standard cell layout designs for integrated circuits while ensuring that the standard cell layout design and manufacturing specifications are met.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a circuit diagram of an integrated circuit, in accordance with some embodiments.

[0005] FIG. 2 is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0006] FIG. 3 is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0007] FIGS. 4A, 4B, 4C, 4D and 4E are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0008] FIG. 5A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0009] FIGS. 5B, 5C and 5D are cross-sectional views of an integrated circuit, in accordance with

some embodiments.

[0010] FIG. 6A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0011] FIGS. 6B, 6C, 6D and 6E are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0012] FIG. 7A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0013] FIGS. 7B, 7C and 7D are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0014] FIG. 8A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0015] FIGS. 8B, 8C, 8D, 8E, 8F and 8G are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0016] FIG. 9A is a diagram of a layout design of an integrated circuit, in accordance with some embodiments.

[0017] FIGS. 9B, 9C, 9D and 9E are cross-sectional views of an integrated circuit, in accordance with some embodiments.

[0018] FIG. 10 is a flowchart of a method of manufacturing an IC, in accordance with some embodiments.

[0019] FIGS. 11A and 11B are a flowchart of a method of generating a layout design of an integrated circuit, in accordance with some embodiments.

[0020] FIG. 12 is a block diagram of a system of designing an IC layout design, in accordance with some embodiments.

[0021] The patent or application file contains drawings/photographs executed in color. Copies of this patent with color drawing(s)/photograph(s) will be provided by the Office upon request and payment of the necessary fee.

DETAILED DESCRIPTION

[0022] The following disclosure provides different embodiments, or examples, for implementing features of the provided subject matter. Specific examples of components, materials, values, steps, arrangements, or the like, are described below to simplify the present disclosure. These are, of course, merely examples and are not limiting. Other components, materials, values, steps, arrangements, or the like, are contemplated. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0023] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0024] In accordance with some embodiments, an integrated circuit includes a set of active regions in a substrate, a first set of conductive structures, a shallow trench isolation (STI) region, a set of gates and a first set of vias. In some embodiments, the first set of conductive structures are located below the set of gates and the metal-0 (M0) layer of the integrated circuit. In some embodiments,

the first set of conductive structures are referred to as “a buried track,” as the conductive structures of the first set of conductive structures are buried in the substrate, and beneath at least the M0-layer, the MD layer or the POLY layer. In some embodiments, by using the buried track beneath the M0-layer, one M0 track is reduced yielding a standard cell with a smaller height and smaller area compared with other approaches. In some embodiments, the height of a standard cell of the present disclosure is reduced by up to 50% compared with other approaches.

[0025] In some embodiments, the first set of conductive structures includes a single conductive structure that extends in a first direction or a second direction. In some embodiments, the single conductive structure extends in only a single direction (e.g., first direction or second direction) and is referred to as a one directional (1D) structure. In some embodiments, the first set of conductive structures includes a first and a second conductive structure. In some embodiments, the first conductive structure extends in the first direction, and the second conductive structure extends in the second direction, and the first and the second conductive structure are referred to as a two directional (2D) structure.

[0026] In some embodiments, the STI region is between the set of active regions and the first set of conductive structures. In some embodiments, a top surface of the set of conductive structures is below a top surface of the active region of the integrated circuit or a top surface of a fin of the integrated circuit. In some embodiments, the integrated circuit comprises a transmission gate.

[0027] In some embodiments, by utilizing the first set of conductive structures of the integrated circuit of the present disclosure the gate density of a standard cell is increased by up to 30% by offering more routing flexibility than other approaches. In some embodiments, the first set of conductive structures increases routing resources for standard cells compared to other approaches. In some embodiments, the current resistance (IR) and Electromigration (EM) of the present disclosure are improved compared to other approaches.

[0028] FIG. 1 is a circuit diagram of an integrated circuit **100**, in accordance with some embodiments. In some embodiments, integrated circuit **100** corresponds to a transmission gate.

[0029] Integrated circuit **100** includes two P-type metal oxide semiconductor (PMOS) transistors **P1** and **P2**, and two N-type metal oxide semiconductor (NMOS) transistors **N1** and **N2**.

[0030] PMOS transistor **P1** includes a gate terminal **A1**, a terminal **T1** and a terminal **T3**.

[0031] NMOS transistor **N1** includes a gate terminal **B2**, a terminal **T2** and a terminal **T4**.

[0032] Terminal **T1** of PMOS transistor **P1** is electrically coupled to terminal **T2** of NMOS transistor **N1**. Terminal **T3** of PMOS transistor **P1** is electrically coupled to terminal **T4** of NMOS transistor **N1**. NMOS transistor **N1** and PMOS transistor **P1** are coupled to each other to form a first transistor pair.

[0033] PMOS transistor **P2** includes a gate terminal **B1**, a terminal **T5** and a terminal **T7**.

[0034] NMOS transistor **N2** includes a gate terminal **A2**, a terminal **T6** and a terminal **T8**.

[0035] Terminal **T7** of PMOS transistor **P2** is electrically coupled to terminal **T8** of NMOS transistor **N2**. Terminal **T5** of PMOS transistor **P2** is electrically coupled to the terminal **T6** of NMOS transistor **N2**. NMOS transistor **N2** and PMOS transistor **P2** are coupled to each other to form a second transistor pair.

[0036] The gate terminal **A1** of PMOS transistor **P1** is electrically coupled to the gate terminal **A2** of NMOS transistor **N2**. The gate terminal **B1** of PMOS transistor **P2** is electrically coupled to the gate terminal **B2** of NMOS transistor **N1**. Terminal **T3** of PMOS transistor **P1** and terminal **T4** of NMOS transistor **N1** are electrically coupled to the terminal **T6** of NMOS transistor **N2** and the terminal **T5** of PMOS transistor **P2**.

[0037] In some embodiments, at least one terminal of terminals **T1**, **T2**, **T3**, **T4**, **T5**, **T6**, **T7** or **T8** corresponds to a source terminal. In some embodiments, at least one terminal of terminals **T1**, **T2**, **T3**, **T4**, **T5**, **T6**, **T7** or **T8** corresponds to a drain terminal. Other configurations of terminals in integrated circuit **100** are within the scope of the present disclosure.

[0038] In some embodiments, at least integrated circuit **100**, integrated circuit **400** (FIGS. 4A-4E),

layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A) corresponds to one or more standard cells. In some embodiments, the standard cell is a transmission gate. In some embodiments, a standard cell is a logic gate cell. In some embodiments, a logic gate cell includes an AND, OR, NAND, NOR, XOR, INV, AND-OR-Invert (AOI), OR-AND-Invert (OAI), MUX, Flip-flop, BUFF, Latch, delay, clock cells, or the like. In some embodiments, a standard cell is a memory cell. In some embodiments, a memory cell includes a static random access memory (SRAM), a dynamic RAM (DRAM), a resistive RAM (RRAM), a magnetoresistive RAM (MRAM), a read only memory (ROM), or the like. In some embodiments, a standard cell includes one or more active or passive elements. Examples of active elements include, but are not limited to, transistors, diodes, or the like. Examples of transistors include, but are not limited to, metal oxide semiconductor field effect transistors (MOSFET), complementary metal oxide semiconductor (CMOS) transistors, bipolar junction transistors (BJT), high voltage transistors, high frequency transistors, p-channel and/or n-channel field effect transistors (PFETs/NFETs), FinFETs, planar MOS transistors with raised source/drain, or the like. Examples of passive elements include, but are not limited to, capacitors, inductors, fuses, resistors or the like. In some embodiments, IC **100** includes other features not shown for ease of illustration. [0039] FIG. 2 is a diagram of a layout design **200** of an integrated circuit, in accordance with some embodiments.

[0040] Layout design **200** is usable to manufacture integrated circuit **100** of FIG. 1, integrated circuit **400** of FIGS. 4A-4E, integrated circuit **500B** of FIGS. 5B-5D, integrated circuit **600B** of FIGS. 6B-6E, integrated circuit **700B** of FIGS. 7B-7D, integrated circuit **800B** of FIGS. 8B-8G or integrated circuit **900B** of FIGS. 9B-9E. Components that are the same or similar to those in each of FIG. 2-3 or 5A-9E are given the same reference numbers, and detailed description thereof is thus omitted.

[0041] Layout design **200** includes active region layout patterns **202a**, **202b** (collectively referred to as a “set of active region layout patterns **202**”) extending in a first direction X. Active region layout patterns **202a**, **202b** of the set of active region layout patterns **202** are separated from one another in a second direction Y different from the first direction X. The set of active region layout patterns **202** is usable to manufacture a corresponding set of active regions **402** (FIGS. 4A-4E) of integrated circuit **400**. In some embodiments, the set of active region layout patterns **202** is located on a first layout level.

[0042] Layout design **200** further includes gate layout patterns **205a**, **205b**, **205c**, **205d** (collectively referred to as a “set of gate layout patterns **205**”) and gate layout patterns **206a**, **206b**, **206c**, **206d** (collectively referred to as a “set of gate layout patterns **206**”) each extending in the second direction Y. Each of the layout patterns of the set of gate layout patterns **205** is separated from an adjacent layout pattern of the set of gate layout patterns **205** in the first direction X by a first pitch. Each of the layout patterns of the set of gate layout patterns **206** is separated from an adjacent layout pattern of the set of gate layout patterns **206** in the first direction X by the first pitch. Set of gate layout patterns **205** and **206** are separated from each other in the second direction Y. In some embodiments, the set of gate layout patterns **205** and **206** are not separated from each other in the second direction Y forming a single set of gate layout patterns **204** (as shown in FIGS. 3 and 5A-9E). The set of gate layout patterns **204**, **205** or **206** is usable to manufacture a corresponding set of gates **404** (FIGS. 4A-4E) of integrated circuit **400**. The set of gate layout patterns **204**, **205** and **206** are positioned on a second layout level (POLY) different from the first layout level. The set of active region layout patterns **202** is below the set of gate layout patterns **204**, **205** or **206**.

[0043] Gate layout pattern **206b** is usable to manufacture the gate terminal A1 of PMOS transistor P1 of FIG. 1. Gate layout pattern **206c** is usable to manufacture the gate terminal B1 of PMOS transistor P2 of FIG. 1. Gate layout pattern **205b** is usable to manufacture the gate terminal B2 of

NMOS transistor N1 of FIG. 1. Gate layout pattern **205c** is usable to manufacture the gate terminal **A2** of NMOS transistor N2 of FIG. 1.

[0044] Layout design **200** further includes metal over diffusion layout patterns **210a**, **210b**, **210c** (collectively referred to as a “set of metal over diffusion layout patterns **210**”) extending in the second direction Y. Each of the layout patterns of the set of metal over diffusion layout patterns **210** is separated from an adjacent layout pattern of the set of metal over diffusion layout patterns **210** in the first direction X. The set of metal over diffusion layout patterns is located on the second layout level. The set of metal over diffusion layout patterns **210** is usable to manufacture a corresponding set of contacts **410** (FIGS. 4A-4E) of integrated circuit **400**.

[0045] In some embodiments, metal over diffusion layout pattern **210a** is usable to manufacture terminal T1 of PMOS transistor P1 of FIG. 1, and terminal T2 of NMOS transistor N1 of FIG. 1. In some embodiments, metal over diffusion layout pattern **210a** is usable to manufacture terminal T7 of PMOS transistor P2 of FIG. 1, and terminal T8 of NMOS transistor N2 of FIG. 1. In some embodiments, metal over diffusion layout pattern **210b** is usable to manufacture terminal T3 of PMOS transistor P1 of FIG. 1, terminal T4 of NMOS transistor N1 of FIG. 1, terminal T5 of PMOS transistor P2 of FIG. 1, and terminal T6 of NMOS transistor N2 of FIG. 1. In some embodiments, metal over diffusion layout pattern **210c** is usable to manufacture terminal T7 of PMOS transistor P2 of FIG. 1, and terminal T8 of NMOS transistor N2 of FIG. 1. In some embodiments, metal over diffusion layout pattern **210c** is usable to manufacture terminal T1 of PMOS transistor P1 of FIG. 1, and terminal T2 of NMOS transistor N1 of FIG. 1.

[0046] Layout design **200** further includes conductive feature layout patterns **212a**, **212b** (collectively referred to as a “set of conductive feature layout patterns **212**”) extending in at least the first direction X or the second direction Y. In some embodiments, the set of conductive feature layout patterns **212** is located on at least the first layout level, the second layout level or a third layout level. In some embodiments, the third layout level is different from the first layout level and the second layout level. The set of conductive feature layout patterns **212** is usable to manufacture a corresponding set of conductive structures (e.g., set of conductive structures **420** or set of conductive structures **430** (FIGS. 4A-4E)) of integrated circuit **400**. In some embodiments, the set of conductive feature layout patterns **212** is usable to manufacture a corresponding set of conductive structures **420'** of integrated circuit **500B**, set of conductive structures **620'** of integrated circuit **600B**, set of conductive structures **720'** of integrated circuit **700B**, set of conductive structures **820'** of integrated circuit **800B** or set of conductive structures **920'** of integrated circuit **900B**. The conductive feature layout pattern **212a** corresponds to fabricating a conductive structure (e.g., conductive structure **420a** or conductive structures **430a**, **430b**, **430c**) used to electrically couple the gate terminal A1 of PMOS transistor P1 to the gate terminal A2 of NMOS transistor N2 of FIG. 1. The conductive feature layout pattern **212b** corresponds to fabricating a conductive structure (e.g., conductive structure **420a** or conductive structures **430a**, **430b**, **430c**) used to electrically couple the gate terminal B1 of PMOS transistor P2 to the gate terminal B2 of NMOS transistor N1 of FIG. 1. In some embodiments, the set of conductive feature layout patterns **212** is not included in layout design **200**.

[0047] FIG. 3 is a diagram of a layout design **300** of an integrated circuit, in accordance with some embodiments. Layout design **300** is usable to manufacture integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E).

[0048] Layout design **300** is a variation of layout design **200** (FIG. 2). Layout design **300** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E).

[0049] In comparison with layout design **200** of FIG. 2, layout design **300** further includes one or more of cut feature layout patterns **304a** and **304b**, fin layout patterns **316a** and **316b**, conductive feature layout patterns **320a**, **330a**, **330b**, **330c** and **340a**, and via layout patterns **322a**, **322b**, **332a**, **332b**, **342a** and **342b**.

[0050] In comparison with layout design **200** of FIG. 2, set of gate layout patterns **204** of layout design **300** replaces set of gate layout patterns **205** and **206**. Set of gate layout patterns **204** corresponds to the set of gate layout patterns **205** and **206** not being separated from each other in the second direction Y thus forming a single set of gate layout patterns (e.g., set of gate layout patterns **204**). The set of gate layout patterns **204** is similar to the set of gate layout patterns **205** and **206**, and similar detailed description is therefore omitted.

[0051] Cut feature layout patterns **304a**, **304b** (collectively referred to as a “set of cut feature layout patterns **304**”) extend in the first direction X. The set of cut feature layout patterns **304** overlaps at least a first gate layout pattern of the set of gate layout pattern **204**. Cut feature layout patterns **304a**, **304b** overlap corresponding gate layout patterns **204b**, **204c** of the set of gate layout patterns **204**. Each layout pattern of the set of cut feature layout patterns **304** has a pattern width in the second direction Y, and a pattern length in the first direction X.

[0052] In some embodiments, cut feature layout pattern **304a** is usable to identify a location of gates **404b1** and **404b2** (FIGS. 4A-4E) of integrated circuit **400** that are removed during operation **1006** of method **1000** (FIG. 10). In some embodiments, cut feature layout pattern **304b** is usable to identify a location of gates **404c1** and **404c2** (FIGS. 4A-4E) of integrated circuit **400** that are removed during operation **1006** of method **1000** (FIG. 10). The removed gate portion of gates **404b1** and **404b2** and gates **404c1** and **404c2** is also referred to as a cut region. In some embodiments, the set of gate layout patterns **204** and the cut feature layout pattern **304** are located on the second layout level. In some embodiments, the second layout level corresponds to a poly layout level (POLY). Other configurations or quantities of patterns in the set of cut feature layout patterns **304** are within the scope of the present disclosure.

[0053] Fin layout patterns **316a**, **316b** (collectively referred to as “set of fin layout patterns **316**”) each extend in the first direction X. Fin layout patterns **316a**, **316b** are over corresponding active region layout pattern **202a**, **202b**. The set of fin layout patterns **316** are below the set of gate layout patterns **204**. Each of the layout patterns of the set of fin layout patterns **316** is separated from an adjacent layout pattern of the set of fin layout patterns **316** in the second direction Y by a fin pitch. The set of fin layout patterns **316** is usable to manufacture a corresponding set of fins of integrated circuit **400**. Other configurations or quantities of fins in the set of fin layout patterns **316** are within the scope of the present disclosure.

[0054] Conductive feature layout pattern **320a** (hereinafter referred to as a “set of conductive feature layout patterns **320**”) extends in at least the first direction X or the second direction Y. In some embodiments, the set of conductive feature layout patterns **320** extends in a single direction (e.g., first direction X or second direction Y). In some embodiments, the set of conductive feature layout patterns **320** extends in two directions (e.g., first direction X and second direction Y (FIG. 5A)). The set of conductive feature layout patterns **320** includes one or more conductive feature layout patterns. The set of conductive feature layout patterns **320** is located on the first layout level. In some embodiments, the first layout level of layout design **300** is the active region. In some embodiments, the first layout level of layout design **300** is divided into a metal under (MU) portion and a via under (VU) portion. In some embodiments, the MU portion is below the VU portion. In some embodiments, the set of conductive feature layout patterns **320** is located on the MU portion of the first layout level. In some embodiments, the MU portion is positioned beneath at least the M0 level, the POLY level or the MD level of layout design **300**.

[0055] The set of conductive feature layout patterns **320** is between the set of active region layout patterns **202**. Conductive feature layout pattern **320a** is positioned between active region layout pattern **202a** and active region layout pattern **202b**. The set of conductive feature layout patterns **320** is usable to manufacture a corresponding set of conductive structures **420** (FIGS. 4A-4E) of integrated circuit **400**. The set of conductive feature layout patterns **320** is below the set of gate layout patterns **204** and the set of metal over diffusion layout patterns **210**. In some embodiments, the set of metal over diffusion layout patterns **210** overlaps the set of conductive feature layout

patterns **320**. The set of gate layout patterns **204** overlaps at least the set of conductive feature layout patterns **320**. Other configurations or quantities of patterns in the set of conductive feature layout patterns **320** are within the scope of the present disclosure.

[0056] Via layout patterns **322a**, **322b** (collectively referred to as a “set of via layout patterns **322**”) are between the set of conductive feature layout patterns **320** and the set of gate layout patterns **204**. Via layout patterns **322a**, **322b** of the set of via layout patterns **322** are between corresponding gate layout pattern **204c**, **204b** of the set of gate layout patterns **204** and the set of conductive feature layout patterns **320**. Via layout patterns **322a**, **322b** are positioned at a VU portion of the first layout level of layout design **300**. In some embodiments, the VU portion is above the MU portion. In some embodiments, the VU portion is positioned beneath at least the V0 level, the VG level or the VD level of layout design **300**.

[0057] The set of via layout patterns **322** is usable to manufacture a corresponding set of vias **422** (FIGS. **4A-4E**). The set of vias **422** couple the set of conductive structures **420** to the set of gates **404**.

[0058] In some embodiments, a via layout pattern **322a**, **322b** of the set of via layout patterns **322** is located where a corresponding gate layout pattern **204c**, **204b** of the set of gate layout patterns **204** overlaps a corresponding layout pattern of the set of conductive feature layout patterns **320**. In some embodiments, a center of via layout pattern **322a**, **322b** of the set of via layout patterns **322** is below a center of a corresponding gate layout pattern **204c**, **204b** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **322a**, **322b** of the set of via layout patterns **322** is aligned in at least the first direction X or the second direction Y with a center of a corresponding gate layout pattern **204c**, **204b** of the set of gate layout patterns **204**. Other configurations of via layout patterns **322a**, **322b** of the set of via layout patterns **322** are within the scope of the present disclosure.

[0059] Conductive feature layout patterns **330a**, **330b**, **330c** (collectively referred to as a “set of conductive feature layout patterns **330**”) extend in the first direction X. The set of conductive feature layout patterns **330** is positioned between the set of active region layout patterns **202**. Each of conductive feature layout patterns **330a**, **330b**, **330c** is positioned between active region layout pattern **202a** and active region layout pattern **202b**. In some embodiments, each of the layout patterns **330a**, **330b**, **330c** of the set of conductive feature layout patterns **330** is separated from an adjacent layout pattern of the set of conductive feature layout patterns **330** in the second direction Y. The set of conductive feature layout patterns **330** is usable to manufacture a corresponding set of conductive structures **430** (FIGS. **4A-4E**) of integrated circuit **400**.

[0060] The set of conductive feature layout patterns **330** is located on a third layout level different from the first layout level and the second layout level. In some embodiments, the third layout level corresponds to the metal zero (M0) layout level. In some embodiments, the third layout level corresponds to an upper metal layer including the metal one (M1) layout level, metal two (M2) layout level, metal three (M3) layout level, or the like. The set of conductive feature layout patterns **330** overlaps at least the set of gate layout patterns **204** or the set of metal over diffusion layout patterns **210**. The set of conductive feature layout patterns **330** is above the set of conductive feature layout patterns **320** and the set of active region layout patterns **202**.

[0061] Via layout patterns **332a**, **332b** (collectively referred to as a “set of via layout patterns **332**”) are between the set of conductive feature layout patterns **330** and the set of gate layout patterns **204**. Via layout patterns **332a**, **332b** are positioned at a via over gate (VG) level of layout design **300**. Via layout patterns **332a**, **332b** of the set of via layout patterns **332** are between a corresponding conductive feature layout pattern **330c**, **330a** of the set of conductive feature layout patterns **330** and a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. The set of via layout patterns **332** is usable to manufacture a corresponding set of vias **432** (FIGS. **4A-4E**). The set of vias **432** couples the set of conductive structures **430** to the set of gates **404**.

[0062] In some embodiments, a via layout pattern **332a**, **332b** of the set of via layout patterns **332** is located where a corresponding layout pattern **330c**, **330a** of the set of conductive feature layout patterns **330** overlaps a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **332a**, **332b** of the set of via layout patterns **332** is over a center of a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **332a**, **332b** of the set of via layout patterns **332** is below a center of a corresponding conductive feature layout pattern **330c**, **330a** of the set of conductive feature layout patterns **330**. In some embodiments, a center of via layout pattern **332a**, **332b** of the set of via layout patterns **332** is aligned in at least the first direction X or the second direction Y with a center of a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **332a**, **332b** of the set of via layout patterns **332** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **330c**, **330a** of the set of conductive feature layout patterns **330**. Other configurations of via layout patterns **332a**, **332b** of the set of via layout patterns **332** are within the scope of the present disclosure.

[0063] Conductive feature layout pattern **340a** (hereinafter referred to as a “set of conductive feature layout patterns **340**”) extends in the second direction Y. The set of conductive feature layout patterns **340** includes one or more conductive feature layout patterns. The set of conductive feature layout patterns **340** is between the set of gate layout patterns **204**. In some embodiments, conductive feature layout pattern **340a** is positioned between gate layout pattern **204c** and gate layout pattern **204d**. In some embodiments, each of the layout patterns of the set of conductive feature layout patterns **340** is separated from an adjacent layout pattern of the set of conductive feature layout patterns **340** in the first direction X. The set of conductive feature layout patterns **340** is usable to manufacture a corresponding set of conductive structures **440** (FIGS. 4A-4E) of integrated circuit **400**.

[0064] The set of conductive feature layout patterns **340** overlaps at least the set of conductive feature layout patterns **320**, the set of conductive feature layout patterns **330**, the set of metal over diffusion layout patterns **210** or the set of active region layout patterns **202**. The set of conductive feature layout patterns **340** is located on a fourth layout level different from the first layout level, the second layout level and the third layout level. In some embodiments, the fourth layout level corresponds to a metal one (M1) level of layout design **300**. Other configurations or quantities of patterns in the set of conductive feature layout patterns **340** are within the scope of the present disclosure.

[0065] Via layout patterns **342a**, **342b** (collectively referred to as a “set of via layout patterns **342**”) are between the set of conductive feature layout patterns **330** and the set of conductive feature layout patterns **340**. Via layout patterns **342a**, **342b** of the set of via layout patterns **342** are between corresponding conductive feature layout pattern **330a**, **330c** of the set of conductive feature layout patterns **330** and the set of conductive feature layout patterns **340**. Via layout patterns **342a**, **342b** are positioned at a via zero (V0) level of layout design **300**. The set of via layout patterns **342** is usable to manufacture a corresponding set of vias **442** (FIGS. 4A-4E). The set of vias **442** couples the set of conductive structures **430** to the set of conductive structures **440**.

[0066] In some embodiments, a via layout pattern **342a**, **342b** of the set of via layout patterns **342** is located where a corresponding conductive feature layout pattern **330a**, **330c** of the set of conductive feature layout patterns **330** is overlapped by a layout pattern of the set of conductive feature layout patterns **340**. In some embodiments, a center of via layout pattern **342a**, **342b** of the set of via layout patterns **342** is above a center of a corresponding conductive feature layout pattern **330a**, **330c** of the set of conductive feature layout patterns **330**. In some embodiments, a center of via layout pattern **342a**, **342b** of the set of via layout patterns **342** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **330a**, **330c** of the set of conductive feature layout patterns **330**. Other configurations of via

layout patterns **342a**, **342b** of the set of via layout patterns **342** are within the scope of the present disclosure.

[0067] In some embodiments, gate terminal **A1**, **404b1** (FIG. **4C**) (manufactured by gate layout pattern **204b**) of layout design **300** is electrically coupled to gate terminal **A2**, **404c2** (FIG. **4B**) (manufactured by gate layout pattern **204c**). For example, in some embodiments, gate terminal **A1**, **404b1** (manufactured by gate layout pattern **204b**) is electrically connected through via **422b** (FIG. **4C**) (manufactured by via layout pattern **322b**) to conductive structure **420a** (manufactured by conductive feature layout pattern **320a**), and conductive structure **420a** is electrically coupled through via **422a** (FIG. **4B**) (manufactured by via layout pattern **322a**) to gate terminal **A2**, **404c2** (manufactured by gate layout pattern **204c**).

[0068] In some embodiments, gate terminal **B1**, **404c1** (manufactured by gate layout pattern **204c**) of layout design **300** is electrically coupled to gate terminal **B2**, **404b2** (manufactured by gate layout pattern **204b**). For example, in some embodiments, gate terminal **B1**, **404c1** (FIG. **4B**) (manufactured by gate layout pattern **204c**) is electrically connected through via **432b** (manufactured by via layout pattern **332b**) to a conductive structure **430a** (manufactured by conductive feature layout pattern **330a**). In some embodiments, conductive structure **430a** is electrically coupled through via **442a**, **V0** (manufactured by via layout pattern **342a**) to a conductive structure **440a**, **M1** (manufactured by conductive feature layout pattern **340a**). In some embodiments, conductive structure **440a** is electrically coupled through via **442b**, **V0** (manufactured by via layout pattern **342b**) to conductive structure **430c** (manufactured by conductive feature layout pattern **330c**). In some embodiments, conductive structure **430c** is electrically coupled through via **432a** (manufactured by via layout pattern **332a**) to gate terminal **B2**, **404b2** (manufactured by gate layout pattern **204b**).

[0069] In some embodiments, the set of conductive feature layout patterns **320** are located below at least the set of gate layout patterns **304**, the metal-0 (**M0**) level or the **MD** level of layout design **300**. In some embodiments, by using the set of conductive feature layout patterns **320**, at least one upper metal layer track is reduced in the set of conductive feature layout patterns **330** compared to other approaches. In some embodiments, an upper metal layer track includes one or more of **M0**, **M1** or **M2**, or the like. In some embodiments, the set of conductive feature layout patterns **330** includes an upper metal layer track layout pattern located on the **M1** level, **M2** level, or the like. In some embodiments, by using one less upper metal layer track in the set of conductive feature layout patterns **330**, a standard cell with a smaller height and smaller area results compared with other approaches. In some embodiments, since one upper metal layer track in the set of conductive feature layout patterns **330** is reduced, the width of one or more layout patterns in the set of conductive feature layout patterns **330** is increased, yielding a standard cell with better current resistance (**IR**) and Electromigration (**EM**) compared with other approaches. In some embodiments, by utilizing the set of conductive feature layout patterns **320** of the present disclosure, the gate density of the set of gate layout patterns of a standard cell is increased by up to 30% compared with other approaches. In some embodiments, by utilizing the set of conductive feature layout patterns **320** of the present disclosure, layout design **200** offers more routing flexibility and increases routing resources compared to other approaches.

[0070] FIGS. **4A**, **4B**, **4C**, **4D** and **4E** are cross-sectional views of an integrated circuit **400**, in accordance with some embodiments. FIG. **4A** is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane **A-A'**, FIG. **4B** is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane **B-B'**, FIG. **4C** is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane **C-C'**, FIG. **4D** is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane **D-D'**, and FIG. **4E** is a cross-sectional view of an integrated circuit **400** corresponding to layout design **300** as intersected by plane **E-E'**, in accordance with some embodiments. Integrated circuit **400** is manufactured by layout design

300.

[0071] Integrated circuit **400** is manufactured by layout design **300**. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **400** are similar to the structural relationships and configurations of layout design **200** of FIG. 2 and layout design **300** of FIG. 3, and will not be described in FIGS. 4A-4E for brevity.

[0072] Integrated circuit **400** includes active regions **402a**, **402b** (collectively referred to as “set of active regions **402**”) in a substrate **480**. The set of active regions **402** extends in the first direction X and is located on a first level of integrated circuit **400**. Each active region of the set of active regions **402** is separated from each other in the second direction Y. In some embodiments, set of active regions **402** of integrated circuit **400** is referred to as an oxide definition (OD) region which defines the source or drain diffusion regions of integrated circuit **400**. In some embodiments, the first level of integrated circuit **400** is referred to as the Active/Fin level. In some embodiments, active region **402a**, **402b** includes a set of fins (not shown) extending in the first direction and being below the set of gates **404**. In some embodiments, each of the fins (not shown) is separated from an adjacent fin of the set of fins (not shown) in the second direction Y by a fin pitch (not shown). Other quantities or configurations of the set of active region **402** are within the scope of the present disclosure.

[0073] Integrated circuit **400** further includes a conductive structure **420a** (collectively referred to as a “set of conductive structures **420**”) extending in at least the first direction X or the second direction Y. The set of conductive structures **420** is located on the first level of integrated circuit **400**. The set of conductive structures **420** is between the set of active regions **402**. In some embodiments, conductive structure **420a** is positioned between active regions **402a**, **402b**. In some embodiments, a top surface of conductive structure **420a** of the set of conductive structures **420** is below a top surface of active region **402a**, **402b** of the set of active regions **402** or a top surface of a fin (not shown) of the set of fins (not shown).

[0074] In some embodiments, the first level of integrated circuit **400** is the active region of integrated circuit **400**. In some embodiments, the first level of integrated circuit **400** is divided into a metal under (MU) region and a via under (VU) region. In some embodiments, the MU region is below the VU region. In some embodiments, the set of conductive structures **420** is in the MU region of the first level of integrated circuit **400**. In some embodiments, the conductive structures of the set of conductive structures **420** are buried inside the substrate **480**, and positioned beneath at least the M0 layer, the POLY layer, the MD layer of integrated circuit **400**. In some embodiments, each conductive structure **420a** in the set of conductive structures **420** has a width (not shown) in the second direction Y ranging from about 1-50 nm. In some embodiments, each conductive structure **420a** in the set of conductive structures **420** has a length (not shown) in the first direction X ranging from about 1-10 μm . Other quantities or configurations of the set of conductive structures **404** are within the scope of the present disclosure.

[0075] Integrated circuit **400** further includes a shallow trench isolation (STI) region **490** extending in at least the first direction X or the second direction Y. STI region **490** is located on at least the first level of integrated circuit **400**. In some embodiments, STI region **490** is positioned between the set of active regions **402** and the set of conductive structures **420**. In some embodiments, conductive structure **420a** is separated from each of active regions **402a**, **402b** by STI region **490**. Other quantities or configurations of the STI region **490** are within the scope of the present disclosure.

[0076] Integrated circuit **400** further includes gates **404b1**, **404b2**, **404c1**, **404c2** (collectively referred to as “set of gates **404**”) extending in the second direction Y, overlapping at least the set of conductive structures **420** and being located on a second level of integrated circuit **400**. In some embodiments, the second level of integrated circuit **400** is different from the first level. Each of the gates of the set of gates **404** is separated from an adjacent gate of the set of gates **404** in the first direction X by a first pitch. In some embodiments, the second level of integrated circuit **400** is

referred to as the Poly level. Other quantities or configurations of the set of gates **404** are within the scope of the present disclosure.

[0077] Integrated circuit **400** further includes contacts **410a**, **410b**, **410c** (collectively referred to as a “set of contacts **410**”) extending in the second direction Y, overlapping the set of conductive structures **420**, and being located on the second level of integrated circuit **400**. Each of the contacts **410a**, **410b**, **410c** of the set of contacts **410** is separated from an adjacent contact of the set of contacts **410** in the first direction X. In some embodiments, the set of contacts **410** is located on the MD level of integrated circuit **400**. In some embodiments, at least a contact of the set of contacts **410** is electrically coupled to at least a source or a drain of the active regions **402a**, **402b**. Other quantities or configurations of the set of contacts **410** are within the scope of the present disclosure.

[0078] Integrated circuit **400** further includes vias **422a**, **422b** (collectively referred to as a “set of vias **422**”) coupling the set of conductive structures **420** to the set of gates **404**. The set of vias **422** is between the set of conductive structures **420** and the set of gates **404**. Via **422b** is located where gates **404b1**, **404b2** overlap conductive structure **420a**. Via **422a** is located where gates **404c1**, **404c2** overlap conductive structure **420a**. In some embodiments, set of vias **422** is in the VU level of integrated circuit **400**. The VU level of integrated circuit **400** is between the first level and the second level. Other quantities or configurations of the set of vias **422** are within the scope of the present disclosure.

[0079] Integrated circuit **400** further includes conductive structures **430a**, **430b**, **430c** (collectively referred to as a “set of conductive structures **430**”) extending in the first direction X, being between the set of active regions **402**, overlapping at least the set of gates **404**, and being located on a third level of integrated circuit **400**. In some embodiments, the third level of integrated circuit **400** is different from the first level and the second level. In some embodiments, the third level of integrated circuit **400** is referred to as the M0 level. Each of the conductive structures **430a**, **430b**, **430c** of the set of conductive structures **430** is separated from an adjacent conductive structure of the set of conductive structures **430** in the second direction Y. Other quantities or configurations of the set of conductive structures **430** are within the scope of the present disclosure.

[0080] Integrated circuit **400** further includes vias **432a**, **432b** (collectively referred to as a “set of vias **432**”) coupling the set of conductive structures **430** to the set of gates **404**. The set of vias **432** is between the set of conductive structures **430** and the set of gates **404**. Via **432a** is located where conductive structure **430c** overlaps gates **404b1**, **404b2**. Via **432b** is located where conductive structure **430a** overlaps gates **404c1**, **404c2**. In some embodiments, set of vias **432** is in the VG level of integrated circuit **400**. The VG level of integrated circuit **400** is between the second level and the third level. Other quantities or configurations of the set of vias **432** are within the scope of the present disclosure.

[0081] Integrated circuit **400** further includes conductive structure **440a** (collectively referred to as “set of conductive structures **440**”) extending in the second direction Y, being between the set of gates **404**, overlapping at least the set of conductive structures **420**, and being located on a fourth level of integrated circuit **400**. In some embodiments, the fourth level of integrated circuit **400** is different from the first level, the second level and the third level. In some embodiments, the fourth level of integrated circuit **400** is referred to as the M1 level. In some embodiments, each of the structures of the set of conductive structures **440** is separated from an adjacent structure of the set of conductive structures **440** in the first direction X. Other quantities or configurations of the set of conductive structures **440** are within the scope of the present disclosure.

[0082] Integrated circuit **400** further includes vias **442a**, **442b** (collectively referred to as “set of vias **442**”) coupling the set of conductive structures **440** to the set of conductive structures **430**. The set of vias **442** is between the set of conductive structures **440** and the set of conductive structures **430**. Via **442a** is located where conductive structure **440a** overlaps conductive structure **430a**. Via **442b** is located where conductive structure **440a** overlaps conductive structure **430c**. In some embodiments, set of vias **442** are in the V0 level of integrated circuit **400**. The V0 level of

integrated circuit **400** is between the third level and the fourth level. Other quantities or configurations of the set of vias **442** are within the scope of the present disclosure.

[0083] In some embodiments, one or more conductive structure of the set of conductive structures **420**, **430** or **440** includes a metal, a metal compound or a doped semiconductor. In some embodiments, one or more vias of the set of vias **422**, **432**, **442** includes a metal, a metal compound or a doped semiconductor. In some embodiments, a metal includes at least Cu (Copper), Co, W, Ru, Al, or the like. In some embodiments, a metal compound includes at least AlCu, W—TiN, TiSix, NiSix, TiN, TaN, or the like. In some embodiments, a doped semiconductor includes at least doped silicon, or the like.

[0084] In some embodiments, gate terminal A1, **404b1** is electrically coupled to gate terminal A2, **404c2**. For example, gate terminal A1, **404b1** is electrically connected through via **422b** to conductive structure **420a**, and conductive structure **420a** is electrically coupled through via **422a** to gate terminal A2, **404c2**.

[0085] In some embodiments, gate terminal B1, **404c1** is electrically coupled to gate terminal B2, **404b2**. For example, in some embodiments, gate terminal B1, **404c1** is electrically connected through via **432b** to conductive structure **430a**. In some embodiments, conductive structure **430a** is electrically coupled through via **442a**, V0 to conductive structure **440a**, M1. In some embodiments, conductive structure **440a** is electrically coupled through via **442b**, V0 to conductive structure **430c**. In some embodiments, conductive structure **430c** is electrically coupled through via **432a** to gate terminal B2, **404b2**.

[0086] In some embodiments, the set of conductive structures **420** is located below at least the set of gates **404**, the metal 0 (M0) layer or the MD layer of integrated circuit **400**. In some embodiments, the set of conductive structures **420** is referred to as a buried metal track (BMT), as the conductive structures of the set of conductive structures **420** are buried inside the substrate **480**, and positioned beneath at least the M0 layer, the POLY layer, the MD layer of integrated circuit **400**.

[0087] In some embodiments, by using the set of conductive structures **420**, one upper metal layer track (e.g., MD track, M0 track, M1 track, or the like) is reduced in the set of conductive structures **430** yielding a standard cell with a smaller height and smaller area compared with other approaches. In some embodiments, by using the set of conductive structures **420**, one upper metal layer track (e.g., MD track, M0 track, M1 track, or the like) is reduced in the set of conductive structures **430**, and the space occupied by the removed upper layer metal track can be utilized by increasing a width of at least one conductive structure in the set of conductive structures **430**, yielding a standard cell with better current resistance (IR) and EM compared with other approaches.

[0088] In some embodiments, by reducing one upper metal layer track of the present disclosure, the cell height can be reduced resulting in an increase in the gate density of a standard cell by up to 30% compared with other approaches. In some embodiments, by reducing one upper metal layer track of the present disclosure, integrated circuit **400** offers more routing flexibility and increases routing resources compared to other approaches.

[0089] FIG. 5A is a diagram of a layout design **500A** of an integrated circuit, in accordance with some embodiments.

[0090] FIGS. 5B, 5C and 5D are cross-sectional views of integrated circuit **500B**, in accordance with some embodiments.

[0091] Layout design **500A** is a variation of layout design **300** (FIG. 3). For example, layout design **500A** illustrates an example of where the MU layer (e.g., set of conductive feature layout patterns **520**) extends in two directions.

[0092] Layout design **500A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E) or **500B**.

500A

[0093] In comparison with layout design **300** of FIG. 3, conductive feature layout patterns **520a**, **520b**, and **520c** of layout design **500A** replace conductive feature layout pattern **320a**.

[0094] Conductive feature layout patterns **520a**, **520b**, **520c** (hereinafter referred to as a “set of conductive feature layout patterns **520**”) are located on the first layout level. The set of conductive feature layout patterns **520** is usable to manufacture a corresponding set of two-dimensional (2D) conductive structures **420** or **420'** of integrated circuit **100**, **400** or **500B**.

[0095] In some embodiments, the set of conductive feature layout patterns **520** is located on the MU portion of the first layout level. The set of conductive feature layout patterns **520** extends in two directions (e.g., first direction X and second direction Y). Conductive feature layout pattern **520a** and **520c** each extend in the first direction X. Conductive feature layout pattern **520b** extends in the second direction Y.

[0096] The set of conductive feature layout patterns **520** is between the set of active region layout patterns **202**. Conductive feature layout pattern **520a** is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330b**. Conductive feature layout pattern **520c** is positioned between conductive feature layout pattern **330b** and conductive feature layout pattern **330c**.

[0097] Conductive feature layout pattern **520b** is positioned between gate layout patterns **204b** and **204c**. In some embodiments, conductive feature layout pattern **520b** is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330c**. In some embodiments, metal over diffusion layout pattern **210a**, **210b**, **210c** overlaps a corresponding conductive feature layout pattern **520a**, **520b**, **520c**.

[0098] The set of gate layout patterns **204** overlaps the set of conductive feature layout patterns **520**. In some embodiments, gate layout patterns **204a**, **204b** overlap conductive feature layout pattern **520a**. In some embodiments, gate layout patterns **204c**, **204d** overlap conductive feature layout pattern **520c**. Other configurations or quantities of patterns in the set of conductive feature layout patterns **520** are within the scope of the present disclosure.

[0099] In comparison with layout design **300** of FIG. 3, via layout patterns **522a**, **522b** of layout design **500A** replace corresponding via layout patterns **322a**, **322b**. The set of via layout patterns **522** is similar to the set of via layout patterns **322**, and similar detailed description is therefore omitted. In some embodiments, the set of via layout patterns **522** is usable to manufacture a corresponding set of vias **422'** (FIGS. 5A-5D) coupled between the set of gates **404** and the set of conductive structures **420'** manufactured by the set of conductive feature layout patterns **520**.

[0100] FIG. 5B is a cross-sectional view of integrated circuit **500B** corresponding to layout design **500A** as intersected by plane F-F', FIG. 5C is a cross-sectional view of integrated circuit **500B** corresponding to layout design **500A** as intersected by plane G-G', and FIG. 5D is a cross-sectional view of integrated circuit **500B** corresponding to layout design **500A** as intersected by plane H-H' in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **500B** are similar to the structural relationships and configurations of layout design **500A** of FIG. 5A, and will not be described in FIGS. 5B-5D for brevity.

[0101] Integrated circuit **500B** is manufactured by layout design **500A**. In some embodiments, integrated circuit **500B** is similar to integrated circuit **400**, and similar detailed description is therefore omitted for brevity.

[0102] In comparison with integrated circuit **400** of FIGS. 4A-4E, set of conductive structures **420'** of integrated circuit **500B** replaces set of conductive structures **420**, and set of vias **422'** of integrated circuit **500B** replaces set of vias **422**.

[0103] In some embodiments, the set of conductive structures **420'** includes one or more of conductive structures **420a'**, **420b'** and **420c'**. In some embodiments, the set of vias **422'** includes one or more of vias **422a'** and **422b'**.

[0104] Conductive structures **420a'** and **420c'** extend in the first direction X, and are separated from

each other in the second direction Y. Conductive structure **420b'** extends in the second direction Y. Conductive structure **420b'** directly contacts conductive structures **420a'** and **420c'**. In some embodiments, conductive structures **420a'**, **420b'** and **420c'** are part of a same continuous conductive structure. In some embodiments, set of conductive structures **420'** is referred to as a 2D conductive structure.

[0105] In some embodiments, gate terminal A1 or gate **404b1** (manufactured by gate layout pattern **204b**) of integrated circuit **500B** is electrically coupled to gate terminal A2 or gate **404c2** (manufactured by gate layout pattern **204c**). For example, in some embodiments, gate terminal A1 or gate **404b1** (manufactured by gate layout pattern **204b**) is electrically connected through via **422b'** (manufactured by via layout pattern **522b**) to the two dimensional (2-D) conductive structure **420a'**, **420b'** and **420c'** (manufactured by conductive feature layout patterns **520a**, **520b** **520c**), and the 2-D conductive structure **420a'**, **420b'** and **420c'** is electrically coupled through via **422a'** (manufactured by via layout pattern **522a**) to gate terminal A2 or gate **404c2** (manufactured by gate layout pattern **204c**).

[0106] In some embodiments, gate terminal B1 or gate **404c1** (manufactured by gate layout pattern **204c**) of integrated circuit **500B** is electrically coupled to gate terminal B2 or gate **404b2** (manufactured by gate layout pattern **204b**) similar to integrated circuit **400** (FIGS. 4A-4E), and similar detailed description is therefore omitted.

[0107] FIG. 6A is a diagram of a layout design **600A** of an integrated circuit **600B**, in accordance with some embodiments.

[0108] FIGS. 6B, 6C, 6D and 6E are cross-sectional views of integrated circuit **600B**, in accordance with some embodiments.

[0109] Layout design **600A** is a variation of layout design **300** (FIG. 3). For example, layout design **600A** illustrates an example of where the gate layout pattern (e.g., gate layout patterns **204b**, **204c**) is continuous, and the MU layer (e.g., a set of conductive feature layout patterns **620**) extends in a single direction.

[0110] Layout design **600A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E) or **600B** (FIGS. 6B-6E).

[0111] In comparison with layout design **300** of FIG. 3, layout design **600A** does not include set of cut feature layout patterns **304**. By not including the set of cut feature layout patterns **304**, layout design **600A** is usable to manufacture integrated circuit **600B** having gate terminal A1 directly coupled to gate terminal A2, and gate terminal B1 directly coupled to gate terminal B2. For ease of illustration, each of layout design **600A** of FIG. 6A, layout design **700A** of FIG. 7A, layout design **800A** of FIG. 8A and layout design **900A** of FIG. 9A do not show set of fin layout patterns **316** of FIG. 3.

[0112] In comparison with layout design **300** of FIG. 3, a set of metal over diffusion layout patterns **610** of layout design **600A** replaces set of metal over diffusion layout patterns **210**, and similar detailed description is therefore omitted. The set of metal over diffusion layout patterns **610** is usable to manufacture a corresponding set of contacts **610'** (FIGS. 6B-6E) of integrated circuit **600B**.

[0113] Set of metal over diffusion layout patterns **610** includes one or more of metal over diffusion layout patterns **610a**, **610b**, **610c**, **612a** or **612c**. Metal over diffusion layout patterns **610a**, **610b**, **610c**, **612a** and **612c** are usable to manufacture corresponding contacts **610a'**, **610b'**, **610c'**, **612a'** and **612c'** (FIGS. 6B-6E) of integrated circuit **600B**.

[0114] In comparison with layout design **300** of FIG. 3, metal over diffusion layout patterns **610a**, **612a** of layout design **600A** replace metal over diffusion layout pattern **210a**, metal over diffusion layout pattern **610b** replaces metal over diffusion layout pattern **210b**, and metal over diffusion layout patterns **610c**, **612c** replace metal over diffusion layout pattern **210c**, and similar detailed description is therefore omitted.

[0115] Metal over diffusion layout patterns **610a**, **612a** are similar to metal over diffusion layout

patterns **210a**. Metal over diffusion layout pattern **610a** is separated or discontinuous from metal over diffusion layout patterns **612a**. In some embodiments, metal over diffusion layout pattern **610a** extends in the second direction Y from an edge of active region layout pattern **202b** to an edge of conductive feature layout pattern **330a**. In some embodiments, metal over diffusion layout pattern **612a** extends in the second direction Y from an edge of active region layout pattern **202a** to an edge of a conductive feature layout pattern **630b**.

[0116] Metal over diffusion layout pattern **610b** is similar to metal over diffusion layout pattern **210b** with the reference numeral increased by 400, and similar detailed description is therefore omitted.

[0117] Metal over diffusion layout patterns **610c**, **612c** are similar to metal over diffusion layout patterns **210c**. Metal over diffusion layout pattern **610c** is separated or discontinuous from metal over diffusion layout patterns **612c**. In some embodiments, metal over diffusion layout pattern **610c** extends in the second direction Y from an edge of active region layout pattern **202b** to an edge of conductive feature layout pattern **630b'**. In some embodiments, metal over diffusion layout pattern **612c** extends in the second direction Y from an edge of active region layout pattern **202a** to an edge of conductive feature layout pattern **330c** or **630b'**.

[0118] In comparison with layout design **300** of FIG. 3, a set of via layout patterns **622**, **632** or **642** of layout design **600A** replaces corresponding set of via layout patterns **322**, **332** or **342**, and similar detailed description is therefore omitted. The set of via layout patterns **622**, **632** or **642** is usable to manufacture a corresponding set of vias **622'**, **632'** or **642'** (FIGS. 6B-6E) of integrated circuit **600B**.

[0119] Set of via layout patterns **622** includes one or more of via layout patterns **622a** or **622b**. Set of via layout patterns **632** includes one or more of via layout patterns **632a** or **632b**. Set of via layout patterns **642** includes one or more of via layout patterns **642a** or **642b**.

[0120] Via layout patterns **622a**, **622b**, **632a**, **632b**, **642a** and **642b** are usable to manufacture a corresponding via **622a'**, **622b'**, **632a'**, **632b'**, **642a'** and **642b'** (FIGS. 6B-6E) of integrated circuit **600B**.

[0121] In comparison with layout design **300** of FIG. 3, via layout patterns **622a**, **622b** of layout design **600A** replace corresponding via layout patterns **322a**, **322b**, via layout patterns **632a**, **632b** replace corresponding via layout patterns **332a**, **332b**, and via layout patterns **642a**, **642b** replace corresponding via layout patterns **342a**, **342b**, and similar detailed description is therefore omitted.

[0122] Via layout patterns **622a**, **622b** are similar to corresponding via layout patterns **322a**, **322b**. Via layout patterns **622a**, **622b** (collectively referred to as a “set of via layout patterns **622**”) are between conductive feature layout pattern **620a** and corresponding metal over diffusion layout patterns **612a**, **610c**. The set of via layout patterns **622** is usable to manufacture a corresponding set of vias **622'** (VU) that couple the set of conductive structures **620'** to the set of contacts **610'**.

[0123] In some embodiments, a via layout pattern **622a**, **622b** of the set of via layout patterns **622** is located where a corresponding layout pattern **612a**, **610c** of the set of metal over diffusion layout patterns **610** overlaps conductive feature layout pattern **620a** of the set of conductive feature layout patterns **620**. In some embodiments, a center of via layout patterns **622a**, **622b** of the set of via layout patterns **622** is below a center of a corresponding metal over diffusion layout pattern **612a**, **610c** of the set of metal over diffusion layout patterns **610**. In some embodiments, a center of via layout patterns **622a**, **622b** of the set of via layout patterns **622** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **612a**, **610c** of the set of metal over diffusion layout patterns **610**. Other configurations of via layout patterns **622a**, **622b** of the set of via layout patterns **622** are within the scope of the present disclosure.

[0124] Via layout patterns **632a**, **632b** (collectively referred to as a “set of via layout patterns **632**”) are similar to corresponding via layout patterns **332a**, **332b**. The set of via layout patterns **632** are between the set of conductive feature layout patterns **630** and the set of gate layout patterns **204**.

Via layout pattern **632a**, **632b** of the set of via layout patterns **632** is between a corresponding conductive feature layout pattern **630b**, **630b'** and a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. The set of via layout patterns **632** is usable to manufacture a corresponding set of vias **632'** (FIGS. 6B-6E). As shown in FIGS. 6B-6E, vias **632a'**, **632b'** couple the corresponding conductive structures **660b1'**, **660b2'** to corresponding gates **604b**, **604c**.

[0125] In some embodiments, a via layout pattern **632a**, **632b** of the set of via layout patterns **632** is located where a corresponding layout pattern **630b**, **630b'** overlaps a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is over a center of a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is below a center of a corresponding conductive feature layout pattern **630b**, **630b'**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is aligned in at least the first direction X or the second direction Y with a center of a corresponding gate layout pattern **204b**, **204c** of the set of gate layout patterns **204**. In some embodiments, a center of via layout pattern **632a**, **632b** of the set of via layout patterns **632** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **630b**, **630b'**. Other configurations of via layout patterns **632a**, **632b** of the set of via layout patterns **632** are within the scope of the present disclosure.

[0126] Via layout patterns **642a**, **642b** are similar to corresponding via layout patterns **342a**, **342b**. Via layout pattern **642a**, **642b** (collectively referred to as “set of via layout patterns **642**”) is between conductive feature layout pattern **640a** and corresponding conductive feature layout patterns **330a**, **330c**. Via layout patterns **642a**, **642b** are usable to manufacture corresponding vias **642a'**, **642b'**. As shown in FIGS. 6B-6E, vias **642a'**, **642b'** couple conductive structure **640a'** to corresponding conductive structures **430a**, **430c**.

[0127] In some embodiments, a via layout pattern **642a**, **642b** of the set of via layout patterns **642** is located where layout pattern **640a** of the set of conductive feature layout patterns **640** overlaps a corresponding conductive feature layout pattern **330a**, **330c**. In some embodiments, a center of via layout patterns **642a**, **642b** of the set of via layout patterns **642** is below a center of conductive feature layout pattern **640a**. In some embodiments, a center of via layout patterns **642a**, **642b** of the set of via layout patterns **642** is above a center of a corresponding conductive feature layout pattern **330a**, **330c**. In some embodiments, a center of via layout patterns **642a**, **642b** of the set of via layout patterns **642** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **330a**, **330c**. Other configurations of via layout patterns **642a**, **642b** of the set of via layout patterns **642** are within the scope of the present disclosure.

[0128] In comparison with layout design **300** of FIG. 3, layout design **600A** further includes via layout patterns **636a**, **636b** (collectively referred to as a “set of via layout patterns **636**”).

[0129] Via layout patterns **636a**, **636b** (collectively referred to as a “set of via layout patterns **636**”) are between the set of conductive feature layout patterns **630** and the set of metal over diffusion layout patterns **610**. Via layout pattern **636a**, **636b** of the set of via layout patterns **636** is between a corresponding conductive feature layout pattern **330a**, **330c** and a corresponding metal over diffusion layout pattern **610a**, **612c**. The set of via layout patterns **636** is usable to manufacture a corresponding set of vias **636'** (e.g., VD). As shown in FIGS. 6B-6E, vias **636a'**, **636b'** couple corresponding conductive structures **430a**, **430c** to corresponding contacts **610a'**, **612c'**.

[0130] In some embodiments, a via layout pattern **636a**, **636b** of the set of via layout patterns **636** is located where a corresponding layout pattern **330a**, **330c** overlaps a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is over a center of a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the

set of via layout patterns **636** is below a center of a corresponding conductive feature layout pattern **330a**, **330c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **636a**, **636b** of the set of via layout patterns **636** is aligned in at least the first direction X or the second direction Y with a center of a corresponding conductive feature layout pattern **330a**, **330c**. Other configurations of via layout patterns **636a**, **636b** of the set of via layout patterns **636** are within the scope of the present disclosure.

[0131] In comparison with layout design **300** of FIG. **3**, conductive feature layout pattern **620a** of layout design **600A** replaces conductive feature layout pattern **320a**, conductive feature layout patterns **630b**, **630b'** replace conductive feature layout pattern **330b**, and conductive feature layout pattern **640a** replaces conductive feature layout pattern **340a**, and similar detailed description is therefore omitted.

[0132] Conductive feature layout pattern **620a** is similar to conductive feature layout pattern **320a** with the reference numerals increased by 300. Conductive feature layout pattern **620a** is usable to manufacture a corresponding conductive structure **620a'** (FIGS. **6B-6E**). Other configurations of conductive feature layout pattern **620a** of the set of conductive feature layout patterns **620** are within the scope of the present disclosure.

[0133] Conductive feature layout patterns **630b**, **630b'** are similar to conductive feature layout pattern **330b** with the reference numerals increased by 300. Conductive feature layout patterns **630b**, **630b'** are usable to manufacture corresponding conductive structures **660b1'**, **660b2'** (FIGS. **6B-6E**). Conductive feature layout pattern **630b** is separated or discontinuous from conductive feature layout pattern **630b'**. In some embodiments, conductive feature layout pattern **630b** extends in the first direction X to an edge of metal over diffusion layout pattern **610b**. In some embodiments, conductive feature layout pattern **630b'** extends in the first direction X to another edge of metal over diffusion layout pattern **610b**. Other configurations of conductive feature layout patterns **630b**, **630b'** of the set of conductive feature layout patterns **630** are within the scope of the present disclosure.

[0134] Conductive feature layout pattern **640a** is similar to conductive feature layout pattern **340a** with the reference numerals increased by 300. Conductive feature layout pattern **640a** is usable to manufacture corresponding conductive structure **440a** (FIGS. **6B-6E**). Conductive feature layout pattern **640a** is positioned between gate layout patterns **204b** and **204c**. Other configurations of conductive feature layout pattern **640a** of a set of conductive feature layout patterns **640** are within the scope of the present disclosure.

[0135] FIG. **6B** is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane I-I', FIG. **6C** is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane J-J', FIG. **6D** is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane K-K', and FIG. **6E** is a cross-sectional view of integrated circuit **600B** corresponding to layout design **600A** as intersected by plane L-L', in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **600B** are similar to the structural relationships and configurations of layout design **600A** of FIG. **6A**, and will not be described in FIGS. **6B-6E** for brevity.

[0136] Integrated circuit **600B** is manufactured by layout design **600A**. In some embodiments, integrated circuit **600B** is similar to integrated circuit **400**, and similar detailed description is therefore omitted for brevity.

[0137] In comparison with integrated circuit **400** of FIGS. **4A-4E**, set of contacts **610'** of integrated circuit **600B** replaces set of contacts **410**, set of vias **622'**, **632'**, **642'** of integrated circuit **600B** replace corresponding set of vias **422**, **432**, **442**, conductive structure **620a'** of integrated circuit **600B** replaces conductive structure **420a**, conductive structure **640a'** of integrated circuit **600B**

replaces conductive structure **440a**, conductive structures **660b1'**, **660b2'** of integrated circuit **600B** replaces conductive structure **430b**, gate **604b** of integrated circuit **600B** replaces gates **404b1** and **404b2**, and gate **604c** of integrated circuit **600B** replaces gates **404c1** and **404c2**, and similar detailed description is therefore omitted for brevity.

[0138] In some embodiments, the set of contacts **610'** includes one or more of contacts **610a'**, **610b'**, **610c'**, **612a'** and **612c'**. In some embodiments, the set of vias **622'** includes one or more of vias **622a'** or **622b'**. In some embodiments, the set of vias **632'** includes one or more of vias **632a'** or **632b'**. In some embodiments, the set of vias **642'** includes one or more of vias **642a'** or **642b'**.

[0139] In comparison with contact **410a** of integrated circuit **400** of FIGS. **4A-4E**, contacts **610a'** and **612a'** are not a single, continuous structure. Contacts **610a'** and **612a'** are separated from each other in the second direction **Y**.

[0140] In comparison with contact **410c** of integrated circuit **400** of FIGS. **4A-4E**, contacts **610c'** and **612c'** are not a single, continuous structure. Contacts **610c'** and **612c'** are separated from each other in the second direction **Y**.

[0141] In comparison with conductive structure **430b** of integrated circuit **400** of FIGS. **4A-4E**, conductive structures **660b1'**, **660b2'** are not a single, continuous structure. In other words, conductive structures **660b1'** and **660b2'** are separated from each other in the first direction **X**.

[0142] In comparison with integrated circuit **400** of FIGS. **4A-4E**, integrated circuit **600B** further includes a set of vias **636'**. In some embodiments, the set of vias **636'** includes one or more of vias **636a'** or **636b'**.

[0143] Set of vias **636'** are similar to set of vias **432**, except the set of vias **636'** are positioned between the set of contacts **610'** and one or more of conductive structures **430a**, **430c**, **660b1'** or **660b2'**, and similar detailed description is therefore omitted for brevity. In other words, the set of vias **636'** (e.g., **VD** level) are above the set of contacts **610'** (e.g., **MD** level), and are configured to couple the set of contacts **610'** to upper metal layers (e.g., **M0** level). For example, vias **636a'**, **636b'** couple corresponding conductive structures **430a**, **430c** to corresponding contacts **610a'**, **612c'**. Other configurations of set of vias **636'** are within the scope of the present disclosure.

[0144] In comparison with gates **404b1** and **404b2** of integrated circuit **400** of FIGS. **4A-4E**, gate **604b** is a continuous structure. In some embodiments, gate terminal **A1** (manufactured by gate layout pattern **204b**) of integrated circuit **600B** is directly coupled to gate terminal **A2** (manufactured by gate layout pattern **204b**) by gate **604b**.

[0145] In comparison with gates **404c1** and **404c2** of integrated circuit **400** of FIGS. **4A-4E**, gate **604c** is a continuous structure. In some embodiments, gate terminal **B1** (manufactured by gate layout pattern **204c**) of integrated circuit **600B** is directly coupled to gate terminal **B2** (manufactured by gate layout pattern **204c**) by gate **604c**.

[0146] In some embodiments, terminal **A1D** corresponds to contact **610a'**, and terminal **B2D** corresponds to contact **612c'**. In these embodiments, terminal **A1D** or contact **610a'** (manufactured by metal over diffusion layout pattern **610a**) of integrated circuit **600B** is electrically coupled to terminal **B2D** or contact **612c'** (manufactured by metal over diffusion layout pattern **612c**). For example, in some embodiments, contact **610a'** (manufactured by metal over diffusion layout pattern **610a**) is electrically connected through via **636a'** (manufactured by via layout pattern **636a**) to conductive structure **430a** (manufactured by conductive feature layout pattern **330a**). In some embodiments, conductive structure **430a** is electrically coupled through via **642a'** (manufactured by via layout pattern **642a**) to conductive structure **640a'** (manufactured by conductive feature layout pattern **640a**). In some embodiments, conductive structure **640a'** is electrically coupled through via **642b'** (manufactured by via layout pattern **642b**) to conductive structure **430c** (manufactured by conductive feature layout pattern **330c**). In some embodiments, conductive structure **430c** is electrically coupled through via **636b'** (manufactured by via layout pattern **636b**) to contact **612c'** (manufactured by metal over diffusion layout pattern **612c**). Thus, terminal **A1D** or contact **610a'** is electrically coupled to terminal **B2D** or contact **612c'**.

[0147] In some embodiments, terminal BID corresponds to contact **610c'**, and terminal A2D corresponds to contact **612a'**. In these embodiments, terminal BID or contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) of integrated circuit **600B** is electrically coupled to terminal A2D or contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). For example, in some embodiments, contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) is electrically connected through via **622b'** (manufactured by via layout pattern **622b**) to conductive structure **620a'** (manufactured by conductive feature layout pattern **620a**). In some embodiments, conductive structure **620a'** is electrically coupled through via **622a'** (manufactured by via layout pattern **622a**) to contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). Thus, terminal BID or contact **610c'** of integrated circuit **600B** is electrically coupled to terminal A2D or contact **612a'**.

[0148] Other configurations of terminals A1D, A2D, B1D or B2D are within the scope of the present disclosure. For example, in some embodiments, terminal A1D corresponds to contact **610c'**, and terminal B2D corresponds to contact **612a'**. For example, in some embodiments, terminal A2D corresponds to contact **610a'**, and terminal BID corresponds to contact **612c'**.

[0149] FIG. 7A is a diagram of a layout design **700A** of an integrated circuit, in accordance with some embodiments.

[0150] FIG. 7B, 7C or 7D are cross-sectional views of integrated circuit **700B**, in accordance with some embodiments.

[0151] Layout design **700A** is a variation of layout design **600A** (FIG. 6A). For example, layout design **700A** illustrates an example of where the MU layer (e.g., a set of conductive feature layout patterns **720**) extends in two directions.

[0152] Layout design **700A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E) or **700B** (FIGS. 7B-7D).

[0153] In comparison with layout design **600A** of FIG. 6A, conductive feature layout patterns **720a**, **720b**, and **720c** of layout design **700A** replace conductive feature layout pattern **620a**, and similar detailed description is therefore omitted.

[0154] Conductive feature layout pattern **720a**, **720b**, **720c** (hereinafter referred to as a “set of conductive feature layout patterns **720**”) is similar to corresponding conductive feature layout patterns **520a**, **520b**, **520c** of FIG. 5A.

[0155] The set of conductive feature layout patterns **720** is usable to manufacture a corresponding set of two-dimensional (2D) conductive structures **720'** of integrated circuit **100** or **700B** (FIGS. 7B-7D).

[0156] In some embodiments, the set of conductive feature layout patterns **720** is located on the MU portion of the first layout level. The set of conductive feature layout patterns **720** extends in two directions (e.g., first direction X and second direction Y). Conductive feature layout pattern **720a** and **720c** each extend in the first direction X. Conductive feature layout pattern **720b** extends in the second direction Y.

[0157] The set of conductive feature layout patterns **720** is between the set of active region layout patterns **202**. Conductive feature layout pattern **720a** is positioned between conductive feature layout pattern **330c** and conductive feature layout pattern **630b**. Conductive feature layout pattern **720c** is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **630b'**.

[0158] Conductive feature layout pattern **720b** is positioned between gate layout patterns **204b** and **204c**. In some embodiments, conductive feature layout pattern **720b** is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330c**. In some embodiments, metal over diffusion layout pattern **612a**, **610b**, **610c** overlaps a corresponding conductive feature layout pattern **720a**, **720b**, **720c**.

[0159] The set of gate layout patterns **204** overlaps the set of conductive feature layout patterns **720**. In some embodiments, gate layout patterns **204a**, **204b** overlap conductive feature layout

pattern **720a**. In some embodiments, gate layout patterns **204c**, **204d** overlap conductive feature layout pattern **720c**. Other configurations or quantities of patterns in the set of conductive feature layout patterns **720** are within the scope of the present disclosure.

[0160] FIG. 7B is a cross-sectional view of integrated circuit **700B** corresponding to layout design **700A** as intersected by plane M-M', FIG. 7C is a cross-sectional view of integrated circuit **700B** corresponding to layout design **700A** as intersected by plane N-N', and FIG. 7D is a cross-sectional view of integrated circuit **700B** corresponding to layout design **700A** as intersected by plane O-O' in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **700B** are similar to the structural relationships and configurations of layout design **700A** of FIG. 7A, and will not be described in FIGS. 7B-7D for brevity.

[0161] Integrated circuit **700B** is manufactured by layout design **700A**. In some embodiments, integrated circuit **700B** is similar to integrated circuit **400** or **600A**, and similar detailed description is therefore omitted for brevity.

[0162] In comparison with integrated circuit **600B** of FIGS. 6B-6E, set of conductive structures **720'** of integrated circuit **700B** replaces set of conductive structures **620'**.

[0163] In some embodiments, the set of conductive structures **720'** includes one or more of conductive structures **720a'**, **720b'** and **720c'**.

[0164] Conductive structures **720a'** and **720c'** extend in the first direction X, and are separated from each other in the second direction Y. Conductive structure **720b'** extends in the second direction Y. Conductive structure **720b'** directly contacts conductive structures **720a'** and **720c'**. In some embodiments, conductive structures **720a'**, **720b'** and **720c'** are part of a same continuous conductive structure. In some embodiments, set of conductive structures **720'** is referred to as a 2D conductive structure.

[0165] In some embodiments, gate terminal A1 (manufactured by gate layout pattern **204b**) of integrated circuit **700B** is directly coupled to gate terminal A2 (manufactured by gate layout pattern **204b**) by gate **604b** (manufactured by gate layout pattern **204b**) similar to integrated circuit **600B** FIGS. 6B-6E, and detailed description is therefore omitted.

[0166] In some embodiments, gate terminal B1 (manufactured by gate layout pattern **204c**) of integrated circuit **700B** is directly coupled to gate terminal B2 (manufactured by gate layout pattern **204c**) by gate **604c** (manufactured by gate layout pattern **204c**) similar to integrated circuit **600B** of FIGS. 6B-6E, and detailed description is therefore omitted.

[0167] In some embodiments, terminal A1D (manufactured by metal over diffusion layout pattern **610a**) of integrated circuit **700B** is electrically coupled to terminal B2D (manufactured by metal over diffusion layout pattern **612c**) similar to integrated circuit **600B** of FIGS. 6B-6E, and detailed description is therefore omitted.

[0168] In some embodiments, terminal BID corresponds to contact **610c'**, and terminal A2D corresponds to contact **612a'**. In these embodiments, terminal BID or contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) of integrated circuit **700B** is electrically coupled to terminal A2D or contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). For example, in some embodiments, contact **610c'** (manufactured by metal over diffusion layout pattern **610c**) is electrically connected through via **622b'** (manufactured by via layout pattern **622b**) to conductive structure **720c'** (manufactured by conductive feature layout pattern **720c**). In some embodiments, conductive structure **720c'** (manufactured by conductive feature layout pattern **720c**) is electrically connected to conductive structure **720a'** (manufactured by conductive feature layout pattern **720a**) by conductive structure **720b'** (manufactured by conductive feature layout pattern **720b**). In some embodiments, conductive structure **720a'** (manufactured by conductive feature layout pattern **720a**) is electrically coupled through via **622a'** (manufactured by via layout pattern **622a**) to contact **612a** (manufactured by metal over diffusion layout pattern **612a**). Thus, terminal BID or contact **610c'** of integrated circuit **700B** is electrically coupled to terminal A2D or contact

612a’.

[0169] FIG. **8A** is a diagram of a layout design **800A** of an integrated circuit, in accordance with some embodiments.

[0170] FIGS. **8B**, **8C**, **8D**, **8E**, **8F** and **8G** are cross-sectional views of integrated circuit **800B**, in accordance with some embodiments.

[0171] Layout design **800A** is a variation of layout design **300** (FIG. **3**) and layout design **700A** (FIG. **7A**). For example, layout design **800A** illustrates an example layout design having a **4** poly pitch width in the first direction X, and the MU layer (e.g., a set of conductive feature layout patterns **820**) extends in a single direction, but includes two portions (e.g., conductive feature layout patterns **820a** and **820b**).

[0172] Layout design **800A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. **1**), integrated circuit **400** (FIGS. **4A-4E**) or integrated circuit **800B** (FIGS. **8B-8G**).

[0173] In comparison with layout design **300** of FIG. **3**, layout design **800A** does not include set of cut feature layout patterns **304**. By not including the set of cut feature layout patterns **304**, layout design **800A** is usable to manufacture an integrated circuit having gate terminal **A1** directly coupled to gate terminal **A2**.

[0174] In comparison with layout design **200**, **300**, **500A**, **600A** or **700A**, layout design **800A** further includes an additional gate layout pattern **804e** extending in the second direction Y, resulting in a wider layout design (e.g., layout design **800A**). For example, layout design **800A** has a **4** poly pitch width in the first direction X. Layout design **800A** has features of layout design **300** and **600A**. For example, layout design incorporates at least metal diffusion layout patterns **610a**, **610c**, **612a** and **612c** from layout design **600A**. Similarly, layout design **800A** incorporates at least metal diffusion layout patterns **210a**, and **210c** from layout design **300**.

[0175] Layout design **800A** includes set of active region layout patterns **202**, gate layout patterns **204a**, **204b**, **204c**, **204d**, **804e** (collectively referred to as a “set of gate layout patterns **804**”), metal over diffusion layout patterns **210a**, **210c**, **610a**, **610c**, **612a**, **612c** (collectively referred to as a “set of metal over diffusion layout patterns **810**”), set of conductive feature layout patterns **330**, conductive feature layout pattern **820a**, **820b** (collectively referred to as a “set of conductive feature layout patterns **820**”), via layout patterns **822a**, **822b**, **822c**, **822d** (collectively referred to as a “set of via layout patterns **822**”), via layout patterns **832a**, **832b**, **832c** (collectively referred to as a “set of via layout patterns **832**”) and via layout patterns **836a**, **836b** (collectively referred to as a “set of via layout patterns **836**”).

[0176] In comparison with layout design **200**, **300**, **500A**, **600A** or **700A**, set of active region layout patterns **202** and set of conductive feature layout patterns **330** extend across at least gate layout patterns **204a** and **804e**.

[0177] Gate layout pattern **804e** is similar to one or more of gate layout patterns **204a**, **204b**, **204c**, **204d** of the set of gate layout patterns **204**, and detailed description is therefore omitted. Gate layout pattern **804e** is separated from gate layout pattern **204d** in the first direction X by a Poly pitch. The set of gate layout patterns **804** is usable to manufacture a corresponding set of gates **804’** (FIGS. **8B-8G**) of integrated circuit **800B**. Gate layout pattern **804e** is usable to manufacture a corresponding gate **804e’** (FIGS. **8B-8G**) of integrated circuit **800B**.

[0178] Set of metal over diffusion layout patterns **810** is similar to set of metal over diffusion layout patterns **210** of FIG. **2** and set of metal over diffusion layout patterns **610** of FIGS. **6A** & **7A**, and detailed description is therefore omitted. The set of metal over diffusion layout patterns **810** is usable to manufacture a corresponding set of contacts **810’** (FIGS. **8B-8G**) of integrated circuit **800B**. Metal over diffusion layout patterns **210a**, **210c**, **610a**, **610c**, **612a** and **612c** are usable to manufacture corresponding contacts **410a**, **410c**, **610a’**, **610c’**, **612a’** and **612c’** (FIGS. **8B-8G**) of integrated circuit **800B**.

[0179] Metal over diffusion layout pattern **210a** is positioned between gate layout patterns **204a**

and **204b**. Metal over diffusion layout pattern **210c** is positioned between gate layout patterns **204d** and **804e**. Each of metal over diffusion layout patterns **610a** and **612a** are positioned between gate layout patterns **204b** and **204c**. Each of metal over diffusion layout patterns **610c** and **612c** are positioned between gate layout patterns **204c** and **204d**.

[0180] Set of conductive feature layout patterns **820** is similar to set of conductive feature layout patterns **320**, **420**, **520**, **620**, **720**, and detailed description is therefore omitted. The set of conductive feature layout patterns **820** is usable to manufacture a corresponding set of conductive structures **820'** (FIGS. **8B-8G**) of integrated circuit **800B**. Conductive feature layout patterns **820a** and **820b** are similar to corresponding conductive feature layout patterns **720a** and **720c**, and detailed description is therefore omitted. Conductive feature layout patterns **820a** and **820b** are usable to manufacture corresponding conductive structures **820a'** and **820b'** (FIGS. **8B-8G**) of integrated circuit **800B**.

[0181] Conductive feature layout pattern **820a** extends in the first direction X, and is positioned between conductive feature layout pattern **330b** and conductive feature layout pattern **330c**. Conductive feature layout pattern **820a** is positioned between gate layout pattern **204a** and gate layout pattern **204c**. Conductive feature layout pattern **820a** is below at least metal over diffusion layout pattern **210a**, **610a** or **612a**. In some embodiments, metal over diffusion layout pattern **210a** or metal over diffusion layout pattern **612a** overlaps conductive feature layout pattern **820a**.

[0182] Conductive feature layout pattern **820b** extends in the first direction X, and is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330b**. Conductive feature layout pattern **820b** is positioned between gate layout pattern **204c** and gate layout pattern **804e**. Conductive feature layout pattern **820b** is below at least metal over diffusion layout pattern **210c**, **610c** or **612c**. In some embodiments, metal over diffusion layout pattern **210c** or metal over diffusion layout pattern **610c** overlaps conductive feature layout pattern **820b**.

[0183] Set of via layout patterns **822** is similar to set of via layout patterns **322**, **522**, **622**, and detailed description is therefore omitted. The set of via layout patterns **822** is usable to manufacture a corresponding set of vias **822'** (FIGS. **8B-8G**) of integrated circuit **800B**.

[0184] Via layout patterns **822a**, **822b**, **822c** and **822d** are usable to manufacture corresponding vias **822a'**, **822b'**, **822c'** and **822d'** (FIGS. **8B-8G**) of integrated circuit **800B**. Via layout patterns **822a**, **822c** is similar to via layout pattern **622a**, and detailed description is therefore omitted. Via layout pattern **822c**, **822a** is between conductive feature layout pattern **820a** and corresponding metal over diffusion layout patterns **210a**, **612a**. In some embodiments, via layout pattern **822c**, **822a** of the set of via layout patterns **822** is located where a corresponding layout pattern **210a**, **612a** of the set of metal over diffusion layout patterns **810** overlaps conductive feature layout pattern **820a** of the set of conductive feature layout patterns **820**. In some embodiments, a center of via layout patterns **822c**, **822a** of the set of via layout patterns **822** is below a center of a corresponding metal over diffusion layout pattern **210a**, **612a** of the set of metal over diffusion layout patterns **810**. In some embodiments, a center of via layout patterns **822c**, **822a** of the set of via layout patterns **822** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **210a**, **612a** of the set of metal over diffusion layout patterns **810**.

[0185] Via layout patterns **822b**, **822d** is similar to via layout pattern **622b**, and detailed description is therefore omitted. Via layout pattern **822b**, **822d** is between conductive feature layout pattern **820b** and corresponding metal over diffusion layout patterns **610c**, **210c**. In some embodiments, a via layout pattern **822b**, **822d** of the set of via layout patterns **822** is located where a corresponding layout pattern **610c**, **210c** of the set of metal over diffusion layout patterns **810** overlaps conductive feature layout pattern **820b** of the set of conductive feature layout patterns **820**. In some embodiments, a center of via layout patterns **822b**, **822d** of the set of via layout patterns **822** is below a center of a corresponding metal over diffusion layout pattern **610c**, **210c** of the set of metal over diffusion layout patterns **810**. In some embodiments, a center of via layout patterns **822b**,

822d of the set of via layout patterns **822** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **610c**, **210c** of the set of metal over diffusion layout patterns **810**. Other configurations of via layout patterns **822a**, **822b**, **822c**, **822d** of the set of via layout patterns **822** are within the scope of the present disclosure. [0186] Set of via layout patterns **832** is similar to set of via layout patterns **332**, **632**, and detailed description is therefore omitted. The set of via layout patterns **832** is usable to manufacture a corresponding set of vias **832'** (FIGS. **8B-8G**) of integrated circuit **800B**. The set of via layout patterns **832** are between the set of conductive feature layout patterns **330** and the set of gate layout patterns **804**. Via layout patterns **832a**, **832b** and **832c** are usable to manufacture corresponding vias **832a'**, **832b'** and **832c'** (FIGS. **8B-8G**) of integrated circuit **800B**. Via layout patterns **832a**, **832c** of the set of via layout patterns **832** are between conductive feature layout pattern **330c** and a corresponding gate layout pattern **204b**, **204d** of the set of gate layout patterns **804**. Via layout pattern **832b** of the set of via layout patterns **832** is between conductive feature layout pattern **330a** and gate layout pattern **204c** of the set of gate layout patterns **804**.

[0187] In some embodiments, a via layout pattern **832a**, **832c** of the set of via layout patterns **832** is located where layout pattern **330c** overlaps a corresponding gate layout pattern **204b**, **204d** of the set of gate layout patterns **804**. In some embodiments, via layout pattern **832b** of the set of via layout patterns **832** is located where layout pattern **330a** overlaps gate layout pattern **204c** of the set of gate layout patterns **804**. In some embodiments, a center of via layout pattern **832a**, **832c** of the set of via layout patterns **832** is over a center of a corresponding gate layout pattern **204b**, **204d** of the set of gate layout patterns **804**. In some embodiments, a center of via layout pattern **832b** of the set of via layout patterns **832** is over a center of gate layout pattern **204c** of the set of gate layout patterns **804**. In some embodiments, a center of via layout pattern **832a**, **832c** of the set of via layout patterns **832** is below a center of conductive feature layout pattern **330c**. In some embodiments, a center of via layout pattern **832b** of the set of via layout patterns **832** is below a center of conductive feature layout pattern **330a**. Other configurations of via layout patterns **832a**, **832b**, **832c** of the set of via layout patterns **832** are within the scope of the present disclosure.

[0188] Set of via layout patterns **836** is similar to set of via layout patterns **636**, and detailed description is therefore omitted. The set of via layout patterns **836** is usable to manufacture a corresponding set of vias **836'** (FIGS. **8B-8G**) of integrated circuit **800B**. Set of via layout patterns **836** are between the set of conductive feature layout patterns **330** and the set of metal over diffusion layout patterns **810**. Via layout patterns **836a** and **836b** are usable to manufacture corresponding vias **836a'** and **836b'** (FIGS. **8B-8G**) of integrated circuit **800B**. Via layout patterns **836a**, **836b** of the set of via layout patterns **836** are between conductive feature layout pattern **330b** and a corresponding metal over diffusion layout pattern **610a**, **612c**.

[0189] In some embodiments, a via layout pattern **836a**, **836b** of the set of via layout patterns **836** is located where layout pattern **330b** at least partially overlaps a corresponding metal over diffusion layout pattern **610a**, **612c**. Via layout pattern **836a**, **836b** of the set of via layout patterns **836** is over a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **836a**, **836b** of the set of via layout patterns **836** is over a center of a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **836a**, **836b** of the set of via layout patterns **836** is below a center of conductive feature layout pattern **330b**. In some embodiments, a center of via layout pattern **836a**, **836b** of the set of via layout patterns **836** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **610a**, **612c**. In some embodiments, a center of via layout pattern **836a**, **836b** of the set of via layout patterns **836** is aligned in at least the first direction X or the second direction Y with a center of conductive feature layout pattern **330b**. Other configurations of via layout patterns **836a**, **836b** of the set of via layout patterns **836** are within the scope of the present disclosure.

[0190] FIG. **8B** is a cross-sectional view of integrated circuit **800B** corresponding to layout design

800A as intersected by plane P-P', FIG. **8C** is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane Q-Q', FIG. **8D** is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane R-R', FIG. **8E** is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane S-S', FIG. **8F** is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane T-T', and FIG. **8G** is a cross-sectional view of integrated circuit **800B** corresponding to layout design **800A** as intersected by plane U-U', in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **800B** are similar to the structural relationships and configurations of layout design **800A** of FIG. **8A**, and will not be described in FIGS. **8B-8G** for brevity.

[0191] Integrated circuit **800B** is manufactured by layout design **800A**. In some embodiments, integrated circuit **800B** is similar to integrated circuit **400**, **600B** or **700B**, and similar detailed description is therefore omitted for brevity.

[0192] Integrated circuit **800B** has a **4** poly pitch width in the first direction X. Integrated circuit **800B** has features of integrated circuit **400** and **600B**. For example, integrated circuit **800B** incorporates at least contacts **610a'**, **610c'**, **612a'** and **612c'** from integrated circuit **600B**. Similarly, integrated circuit **800B** incorporates at least contacts **410a** and **410c** from integrated circuit **400**.

[0193] Integrated circuit **800B** includes set of active regions **402**, gates **404a**, **404d**, **604b**, **604c** and **804e'** (collectively referred to as a set of gates **804'**), contacts **410a**, **410c**, **610a'**, **610c'**, **612a'**, **612c'** (collectively referred to as a set of contacts **810'**), set of conductive structures **430**, conductive structures **820a'**, **820b'** (collectively referred to as a set of conductive structures **820'**), vias **822a'**, **822b'**, **822c'**, **822d'** (collectively referred to as a set of vias **822'**), vias **832a'**, **832b'**, **832c'** (collectively referred to as a set of vias **832'**) and vias **836a'**, **836b'** (collectively referred to as a set of vias **836'**).

[0194] In comparison with integrated circuit **600** (FIGS. **6B-6E**) or **700** (FIGS. **7B-7D**), conductive structures **820a'**, **820b'** replace conductive structures **720a'**, **720c'** of integrated circuit **700B**, vias **822a'**, **822b'**, **822c'**, **822d'** replace vias **622a'**, **622b'** of integrated circuit **700B**, vias **832a'**, **832b'**, **832c'** replace vias **632a'**, **632b'** of integrated circuit **600B**, vias **836a'**, **836b'** replace vias **636a'**, **636b'** of integrated circuit **600B**, and similar detailed description is therefore omitted for brevity.

[0195] In comparison with integrated circuit **600B** (FIGS. **6B-6E**) or **700B** (FIGS. **7B-7D**), gate **804e'** is similar to gate **404a**, **404d**, **604b** or **604c**, and similar detailed description is therefore omitted for brevity.

[0196] In comparison with conductive structures **720a'** and **720c'** of integrated circuit **700B** of FIGS. **7B-7D**, conductive structures **820a'** and **820b'** are not coupled together. In other words, conductive structures **820a'** and **820b'** are not coupled together through another conductive structure (e.g., conductive structure **720b'**).

[0197] The set of vias **822'** electrically couple the set of conductive structures **820'** to the set of contacts **810'**. The set of vias **832'** electrically couple the set of conductive structures **430** to the set of gates **804'**. The set of vias **836'** electrically couple the set of conductive structures **430** to the set of contacts **810'**.

[0198] In some embodiments, terminal **A1** or **A2** corresponds to gate **604c**. Gate **604c** is a continuous structure. In some embodiments, gate terminal **A1** is directly coupled to gate terminal **A2** by gate **604c** (manufactured by gate layout pattern **204c** of layout design **800A**).

[0199] In some embodiments, terminal **B1** corresponds to gate **404d**, and terminal **B2** corresponds to gate **604b**. In these embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **800A**) of integrated circuit **800B** is electrically coupled to gate **604b** (manufactured by gate layout pattern **204b** of layout design **800A**). For example, in some embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **800A**) is electrically connected through via **832c'** (manufactured by via layout pattern **832c**) to conductive structure **430c** (manufactured by

conductive feature layout pattern **330c**), and conductive structure **430c** is electrically coupled through via **832a'** (manufactured by via layout pattern **832a**) to gate **604b** (manufactured by gate layout pattern **204b** of layout design **800A**).

[0200] In some embodiments, terminal B2D corresponds to contact **410a**, and terminal B2D' corresponds to contact **612a'**. In these embodiments, terminal B2D or contact **410a** (manufactured by metal over diffusion layout pattern **210a**) of integrated circuit **800B** is electrically coupled to terminal B2D' or contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). For example, in some embodiments, contact **410a** (manufactured by metal over diffusion layout pattern **210a**) is electrically connected through via **822c'** (manufactured by via layout pattern **822c**) to conductive structure **820a'** (manufactured by conductive feature layout pattern **820a**). In some embodiments, conductive structure **820a'** is electrically coupled through via **822a'** (manufactured by via layout pattern **822a**) to contact **612a'** (manufactured by metal over diffusion layout pattern **612a**). Thus, terminal B2D or contact **410a** is electrically coupled to terminal B2D' or contact **612a'**.

[0201] In some embodiments, terminal BID corresponds to contact **410c**, and terminal B1D' corresponds to contact **610c'**. In these embodiments, terminal BID or contact **410c** (manufactured by metal over diffusion layout pattern **210c**) of integrated circuit **800B** is electrically coupled to terminal B1D' or contact **610c'** (manufactured by metal over diffusion layout pattern **610c**). For example, in some embodiments, contact **410c** (manufactured by metal over diffusion layout pattern **210c**) is electrically connected through via **822d'** (manufactured by via layout pattern **822d**) to conductive structure **820b'** (manufactured by conductive feature layout pattern **820b**). In some embodiments, conductive structure **820b'** is electrically coupled through via **822b'** (manufactured by via layout pattern **822b**) to contact **610c'** (manufactured by metal over diffusion layout pattern **610c**). Thus, terminal BID or contact **410c** is electrically coupled to terminal BID' or contact **610c'**.

[0202] FIG. **9A** is a diagram of a layout design **900A** of an integrated circuit **900B**, in accordance with some embodiments.

[0203] FIGS. **9B**, **9C**, **9D** and **9E** are cross-sectional views of integrated circuit **900B**, in accordance with some embodiments.

[0204] Layout design **900A** is a variation of layout design **800A** (FIG. **8A**). For example, layout design **900A** illustrates an example where the gate (e.g., gate **904b1'** and **904b2'**) manufactured by gate layout patterns **204b** is not continuous, and the MU layer (e.g., a set of conductive feature layout patterns **920**) extends in a single direction, but includes two portions (e.g., conductive feature layout patterns **920a** and **920b**).

[0205] Layout design **900A** is usable to manufacture an integrated circuit similar to integrated circuit **100** (FIG. **1**) or integrated circuit **400** (FIGS. **4A-4E**).

[0206] In comparison with layout design **800A** of FIG. **8A**, layout design **900A** further includes a set of cut feature layout patterns **904** and via layout pattern **932a**. By including the set of cut feature layout patterns **904**, layout design **900A** is usable to manufacture an integrated circuit **900B** (FIGS. **9B-9E**) having gate terminal A1 not being coupled to gate terminal B2.

[0207] Set of cut feature layout patterns **904** is usable to cut gate **204b** to form gates **904b1'** and **904b2'** of integrated circuit **900B** (FIGS. **9B-9E**). Set of cut feature patterns **904** is similar to the set of cut feature patterns **304**, and detailed description is therefore omitted.

[0208] Via layout pattern **932a** is similar to one or more layout patterns of set of via layout patterns **332**, **632** and **832**, and detailed description is therefore omitted. The via layout pattern **932a** is usable to manufacture via **932a'** in the set of vias **932'** (FIGS. **9B-9E**).

[0209] Via layout pattern **932a** of layout design **900A** is similar to via layout pattern **832a**, and detailed description is therefore omitted. Via layout pattern **932a** is between the set of conductive feature layout patterns **330** and the set of gate layout patterns **804**. Via layout pattern **932a** is between conductive feature layout pattern **330a** and gate layout pattern **204b** of the set of gate layout patterns **804**. In some embodiments, via layout pattern **932a** is located where layout pattern

330a overlaps gate layout pattern **204b** of the set of gate layout patterns **804**. In some embodiments, a center of via layout pattern **932a** is over a center of gate layout pattern **204b** of the set of gate layout patterns **804**. In some embodiments, a center of via layout pattern **932a** is below a center of conductive feature layout pattern **330a**. Other configurations of via layout pattern **932a** is within the scope of the present disclosure.

[0210] Layout design **900A** is a variation of layout design **800A** (FIG. **8A**). In comparison with layout design **800A** of FIG. **8A**, layout design **900A** does not include via layout pattern **836b**. In comparison with layout design **800A** of FIG. **8A**, set of metal over diffusion layout patterns **910** of layout design **900A** replaces set of metal over diffusion layout patterns **810**, conductive feature layout patterns **920a**, **920b** (collectively referred to as “set of conductive feature layout patterns **920**”) replace a corresponding conductive feature layout pattern **820a**, **820b**, and via layout patterns **922a**, **922c**, **922d** (collectively referred to as “set of via layout patterns **922**”) replace a corresponding via layout pattern **822a**, **822c**, **822d**, and similar detailed description is therefore omitted.

[0211] The set of metal over diffusion layout patterns **910** is usable to manufacture a corresponding set of contacts **910'** (FIGS. **9B-9E**) of integrated circuit **900B**. Set of metal over diffusion layout patterns **910** includes one or more of metal over diffusion layout patterns **210a**, **910b**, **610c**, **612c**, **910c** and **912c**. Metal over diffusion layout patterns **210a**, **910b**, **610c**, **612c**, **910c** and **912c** are usable to manufacture corresponding contacts **410a**, **910b'**, **610c'**, **612c'**, **910c'** and **912c'** (FIGS. **9B-9E**) of integrated circuit **900B**. Metal over diffusion layout pattern **910b** replaces metal over diffusion layout pattern **610a** and metal over diffusion layout pattern **612a** of layout design **800A**, and similar detailed description is therefore omitted. Metal over diffusion layout patterns **910c** and **912c** replace metal over diffusion layout pattern **210c** of layout design **800A**, and similar detailed description is therefore omitted.

[0212] In some embodiments, metal over diffusion layout pattern **910b** extends in the second direction **Y** from an edge of active region layout pattern **202b** to an edge of active region layout pattern **202a**. In some embodiments, metal over diffusion layout pattern **910c** extends in the second direction **Y** from an edge of active region layout pattern **202b** to an edge of conductive feature layout pattern **330b**. In some embodiments, metal over diffusion layout pattern **912c** extends in the second direction **Y** from an edge of active region layout pattern **202a** to another edge of conductive feature layout pattern **330b**. Other configurations of metal over diffusion layout patterns **210a**, **910b**, **610c**, **612c**, **910c** and **912c** of the set of metal over diffusion layout patterns **910** are within the scope of the present disclosure.

[0213] Set of conductive feature layout patterns **920** is similar to set of conductive feature layout patterns **320**, **420**, **520**, **620**, **720** and **820**, and detailed description is therefore omitted. The set of conductive feature layout patterns **920** is usable to manufacture a corresponding set of conductive structures **920'** (FIGS. **9B-9E**) of integrated circuit **900B**. Conductive feature layout patterns **920a** and **920b** are similar to corresponding conductive feature layout patterns **820a** and **820c**, and detailed description is therefore omitted. Conductive feature layout patterns **920a** and **920b** are usable to manufacture corresponding conductive structures **920a'** and **920b'** (FIGS. **9B-9E**) of integrated circuit **900B**.

[0214] Conductive feature layout pattern **920a** extends in the first direction **X**, and is positioned between conductive feature layout pattern **330b** and conductive feature layout pattern **330c**. Conductive feature layout pattern **920a** is positioned between gate layout pattern **204c** and gate layout pattern **804e**. Conductive feature layout pattern **920a** is below at least metal over diffusion layout patterns **612c** or **912c**. In some embodiments, metal over diffusion layout pattern **612c** or metal over diffusion layout pattern **912c** overlaps conductive feature layout pattern **920a**.

[0215] Conductive feature layout pattern **920b** extends in the first direction **X**, and is positioned between conductive feature layout pattern **330a** and conductive feature layout pattern **330b**. Conductive feature layout pattern **920b** is positioned between gate layout pattern **204b** and gate

layout pattern **204d**. Conductive feature layout pattern **920b** is below at least metal over diffusion layout pattern **910b** or **610c**. In some embodiments, metal over diffusion layout pattern **910b** or metal over diffusion layout pattern **610c** overlaps conductive feature layout pattern **920b**. Other configurations of conductive feature layout patterns **920a**, **920b** of the set of conductive feature layout patterns **920** are within the scope of the present disclosure.

[0216] Set of via layout patterns **922** includes via layout patterns **922a**, **822b**, **922c**, **922d**. Set of via layout patterns **922** is similar to set of via layout patterns **322**, **522**, **622** and **822** and detailed description is therefore omitted. The set of via layout patterns **922** is usable to manufacture a corresponding set of vias **922'** (FIGS. **9B-9E**) of integrated circuit **900B** that couple the set of conductive structures **920'** to the set of contacts **910'**.

[0217] Via layout patterns **922a**, **822b**, **922c** and **922d** are usable to manufacture corresponding vias **922a'**, **822b'**, **922c'** and **922d'** (FIGS. **9B-9E**) of integrated circuit **900B**. Via layout patterns **922a**, **922c** is similar to corresponding via layout patterns **822a**, **822c**, and detailed description is therefore omitted. Via layout pattern **922c**, **922a** is between conductive feature layout pattern **920a** and corresponding metal over diffusion layout patterns **612c**, **912c**. In some embodiments, via layout patterns **922c**, **922a** of the set of via layout patterns **922** is located where a corresponding layout pattern **612c**, **912c** of the set of metal over diffusion layout patterns **910** overlaps conductive feature layout pattern **920a** of the set of conductive feature layout patterns **920**. In some embodiments, a center of via layout patterns **922c**, **922a** of the set of via layout patterns **922** is below a center of a corresponding metal over diffusion layout pattern **612c**, **912c** of the set of metal over diffusion layout patterns **910**. In some embodiments, a center of via layout patterns **922c**, **922a** of the set of via layout patterns **922** is aligned in at least the first direction X or the second direction Y with a center of a corresponding metal over diffusion layout pattern **612c**, **912c** of the set of metal over diffusion layout patterns **910**.

[0218] Via layout patterns **922d** is similar to via layout pattern **622b** or **822d**, and detailed description is therefore omitted. Via layout pattern **922d** is between conductive feature layout pattern **920b** and metal over diffusion layout pattern **910b**. In some embodiments, via layout pattern **922d** of the set of via layout patterns **922** is located where layout pattern **910b** of the set of metal over diffusion layout patterns **910** overlaps conductive feature layout pattern **920b** of the set of conductive feature layout patterns **920**. In some embodiments, a center of via layout pattern **922d** of the set of via layout patterns **922** is below a center of metal over diffusion layout pattern **910b** of the set of metal over diffusion layout patterns **910**. In some embodiments, a center of via layout pattern **922d** of the set of via layout patterns **922** is aligned in at least the first direction X or the second direction Y with a center of metal over diffusion layout pattern **910b** of the set of metal over diffusion layout patterns **910**. Other configurations of via layout patterns **922a**, **922c**, **922d** of the set of via layout patterns **922** are within the scope of the present disclosure.

[0219] FIG. **9B** is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane V-V', FIG. **9C** is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane W-W', FIG. **9D** is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane X-X', and FIG. **9E** is a cross-sectional view of integrated circuit **900B** corresponding to layout design **900A** as intersected by plane Y-Y', in accordance with some embodiments. Structural relationships including alignment, lengths and widths, as well as configurations of integrated circuit **900B** are similar to the structural relationships and configurations of layout design **900A** of FIG. **9A**, and will not be described in FIGS. **9B-9E** for brevity.

[0220] Integrated circuit **900B** is manufactured by layout design **900A**. In some embodiments, integrated circuit **900B** is similar to integrated circuit **400**, **600B**, **700B** or **800B**, and similar detailed description is therefore omitted for brevity.

[0221] Integrated circuit **900B** has a **4** poly pitch width in the first direction X.

[0222] Integrated circuit **900B** includes set of active regions **402**, gates **404a**, **404d**, **904b1'**, **904b2'**,

604c and **804e'** (collectively referred to as a set of gates **904'**), contacts **410a**, **910b'**, **610c'**, **612c'**, **910c'** and **912c'** (collectively referred to as a set of contacts **910'**), set of conductive structures **430**, conductive structures **920a'**, **920b'** (collectively referred to as a set of conductive structures **920'**), vias **922a'**, **822b'**, **922c'**, **922d'** (collectively referred to as a set of vias **922'**), vias **832a'**, **832b'**, **832c'**, **932a'** (collectively referred to as a set of vias **932'**) and via **836a'**.

[0223] In comparison with integrated circuit **800B** (FIGS. **8B-8G**), gates **904b1'** and **904b2'** replace gate **604b**, contacts **910b'** replaces contacts **610a'** and **612a'**, contacts **910c'** and **912c'** replace contact **410c**, conductive structures **920a'** and **920b'** replace corresponding conductive structures **820a'** and **820b'**, vias **922a'**, **922c'** and **922d'** replace corresponding vias **822a'**, **822c'** and **822d'**, and similar detailed description is therefore omitted for brevity.

[0224] In comparison with integrated circuit **800B** (FIGS. **8B-8G**), integrated circuit **900B** further includes via **932a'**. Via **932a'** is similar to via **832a'**, and similar detailed description is therefore omitted for brevity. Via **932a'** electrically couples conductive structures **430a** to gate **904b1'**.

[0225] In comparison with integrated circuit **800B** (FIGS. **8B-8G**), integrated circuit **900B** does not include via **836b'**.

[0226] The set of vias **932'** electrically couple the set of conductive structures **430** to the set of gates **904'**. The set of vias **922'** electrically couple the set of conductive structures **920'** to the set of contacts **910'**.

[0227] In some embodiments, terminal **A1** corresponds to gate **904b1'**, terminal **B2** corresponds to gate **904b2'** and terminal **A2** corresponds to gate **604c**. In these embodiments, gate **904b1'** (manufactured by gate layout pattern **204b** of layout design **900A**) is not directly coupled to gate **904b2'** (manufactured by gate layout pattern **204b** of layout design **900A**) because of cut feature layout pattern **904** of layout design **900A**. In other words, in some embodiments, cut feature layout pattern **904** of layout design **900A** is used to identify a removed portion of gate **604b** (shown in FIGS. **8F-8G**) thereby forming discontinuous gates **904b1'** and **904b2'**.

[0228] In some embodiments, terminal **A1** corresponds to gate **904b1'** and terminal **A2** corresponds to gate **604c**. In these embodiments, gate **904b1'** (manufactured by gate layout pattern **204b** of layout design **900A**) of integrated circuit **900B** is electrically coupled to gate **604c** (manufactured by gate layout pattern **204c** of layout design **900A**). For example, in some embodiments, gate **904b1'** (manufactured by gate layout pattern **204b** of layout design **900A**) is electrically connected through via **932a'** (manufactured by via layout pattern **932a**) to conductive structure **430a** (manufactured by conductive feature layout pattern **330a**), and conductive structure **430a** is electrically coupled through via **832b'** (manufactured by via layout pattern **832b**) to gate **604c** (manufactured by gate layout pattern **204c** of layout design **900A**).

[0229] In some embodiments, terminal **B1** corresponds to gate **404d** and terminal **B2** corresponds to gate **904b2'**. In these embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **900A**) is electrically coupled to gate **904b2'** (manufactured by gate layout pattern **204b** of layout design **900A**). For example, in some embodiments, gate **404d** (manufactured by gate layout pattern **204d** of layout design **900A**) is electrically connected through via **832c'** (manufactured by via layout pattern **832c**) to conductive structure **430c** (manufactured by conductive feature layout pattern **330c**), and conductive structure **430c** is electrically coupled through via **832a'** (manufactured by via layout pattern **832a**) to gate **904b2'** (manufactured by gate layout pattern **204b** of layout design **900A**).

[0230] In some embodiments, terminal **A2D** corresponds to contact **612c'**, and terminal **A2D'** corresponds to contact **912c'**. In these embodiments, terminal **A2D** or contact **612c'** (manufactured by metal over diffusion layout pattern **612c** of layout design **900A**) is electrically coupled to terminal **A2D'** or contact **912c'** (manufactured by metal over diffusion layout pattern **912c** of layout design **900A**). For example, in some embodiments, contact **612c'** (manufactured by metal over diffusion layout pattern **612c**) is electrically connected through via **922c'** (manufactured by via layout pattern **922c**) to conductive structure **920a'** (manufactured by conductive feature layout

pattern **920a**). In some embodiments, conductive structure **920a'** (manufactured by conductive feature layout pattern **920a**) is electrically coupled through via **922a'** (manufactured by via layout pattern **922a**) to contact **912c'** (manufactured by metal over diffusion layout pattern **912c** of layout design **900A**). Thus, terminal A2D or contact **612c'** is electrically coupled to terminal A2D' or contact **912c'**.

[0231] In some embodiments, terminal A1D corresponds to contact **910b'**, and terminal A1D' corresponds to contact **610c'**. In these embodiments, terminal A1D or contact **910b'** (manufactured by metal over diffusion layout pattern **910b** of layout design **900A**) is electrically coupled to terminal A1D' or contact **610c'** (manufactured by metal over diffusion layout pattern **610c** of layout design **900A**). For example, in some embodiments, contact **910b'** (manufactured by metal over diffusion layout pattern **910b**) is electrically connected through via **922d'** (manufactured by via layout pattern **922d**) to conductive structure **920b'** (manufactured by conductive feature layout pattern **920b** of layout design **900A**). In some embodiments, conductive structure **920b'** is electrically coupled through via **822b'** (manufactured by via layout pattern **822b**) to contact **610c'** (manufactured by metal over diffusion layout pattern **610c**). Thus, terminal A1D or contact **910b'** is electrically coupled to terminal A1D' or contact **610c'**.

[0232] In some embodiments, the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920** are located below the set of gate layout patterns **304** and the metal-0 (M0) level of layout design **300** & **500A-900A**.

[0233] In some embodiments, by using the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920**, one M0 track is reduced in the set of conductive feature layout patterns **330** of FIGS. 3 & 5A-9A, yielding a standard cell with a smaller height and smaller area compared with other approaches. In some embodiments, by using the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920**, the width of one or more layout patterns in the set of conductive feature layout patterns **330** of FIGS. 3 & 5A-9A is increased since one M0 track is reduced, yielding a standard cell with better current resistance (IR) and Electromigration (EM) compared with other approaches.

[0234] In some embodiments, by utilizing the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920** of the present disclosure, the gate density of the set of gate layout patterns of a standard cell is increased by up to 30% compared with other approaches. In some embodiments, by utilizing the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** and **920** of the present disclosure, layout design **200-300** and **500A-900A** and integrated circuits **400** and **500B-900B** of FIGS. 2-3 & 5A-9E offer more routing flexibility and increases routing resources compared to other approaches.

[0235] FIG. 10 is a flowchart of a method **1000** of forming an integrated circuit in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **1000** depicted in FIG. 10, and that some other processes may only be briefly described herein. In some embodiments, the method **1000** is usable to form integrated circuits, such as integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E), integrated circuit **500B**, integrated circuit **600B**, integrated circuit **700B**, integrated circuit **800B** or integrated circuit **900B**.

[0236] In operation **1002** of method **1000**, a layout design **300** (FIG. 3) of an integrated circuit (e.g., integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E)) is generated. Operation **1002** is performed by a processing device (e.g., processor **1202** (FIG. 12)) configured to execute instructions for generating a layout design (e.g., layout design **200** or **300**). In some embodiments, layout design **200** or **300** is a graphic database system (GDSII) file format. In some embodiments, operation **1002** of method **1000** is utilized to generate one or more of layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A).

[0237] Method **1000** continues with operation **1004**, where the integrated circuit (e.g., integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E)) is manufactured based on the layout design **300** (FIG. 3). In some embodiments, operation **1004** includes manufacturing integrated circuit **500B**, **600B**, **700B**, **800B** or **900B** based on corresponding layout design **500A**, **600A**, **700A**, **800A** or **900A**. In some embodiments, integrated circuit **100** (FIG. 1) or integrated circuit **400** (FIGS. 4A-4E) includes gate **404b1**, **404b2**, **404c1** or **404c2**. In some embodiments, operation **1004** comprises manufacturing at least one mask based on one or more of layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A), and manufacturing the integrated circuit (e.g., integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E), or integrated circuit **500B**, **600B**, **700B**, **800B** or **900B**) based on the at least one mask. In some embodiments, operation **1004** of method **1000** is utilized to manufacture integrated circuit (e.g., integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E), or integrated circuit **500B**, **600B**, **700B**, **800B** or **900B**) based on one or more of layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A).

[0238] Method **1000** continues with operation **1006**, where a portion of a gate (e.g., gate **A1**, **A2**, **B1** or **B2**) is removed thereby forming a first gate structure (e.g., gate **404b1** or **404c1**) and a second gate structure (e.g., gate **404b2** or **404c2**) of integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**. In some embodiments, the portion of the gate (e.g., gate **A1**, **A2**, **B1** or **B2**) that is removed in operation **1006** is identified in layout designs **300**, **500A** and **900A** by cut feature layout pattern **304a** or **304b** (FIGS. 3 and 5A) or cut feature layout pattern **904** (FIG. 9A). In some embodiments, operation **1006** of method **1000** is referred to as a cut-poly (CPO) process. In some embodiments, operation **1006** results in the formation of integrated circuit **100** (FIG. 1), **400** (FIGS. 4A-4E), **500B**, **600B**, **700B**, **800B** or **900B**. In some embodiments, the removed gate portion of gates **404b1** and **404b2** and gates **404c1** and **404c2** is referred to as a cut region.

[0239] In some embodiments, operation **1006** is performed by a removal process. In some embodiments, the removal process includes one or more etching processes suitable to remove a portion of gate (e.g., gate **A1**, **A2**, **B1** or **B2**). In some embodiments, the etching process of operation **1006** includes identifying a portion of the gate (e.g., gate **A1**, **A2**, **B1** or **B2**) that is to be removed, and etching the portion of the gate (e.g., gate **A1**, **A2**, **B1** or **B2**) that is to be removed. In some embodiments, a mask is used to specify portions of the gate (e.g., gate **A1**, **A2**, **B1** or **B2**) that are to be cut or removed. In some embodiments the mask is a hard mask. In some embodiments, the mask is a soft mask. In some embodiments, etching corresponds to plasma etching, reactive ion etching, chemical etching, dry etching, wet etching, other suitable processes, any combination thereof, or the like.

[0240] In some embodiments, method **1000** is utilized with layout designs other than layout design **200** or **300**. For example, in some embodiments, layout design **200** (FIG. 2) or **300** (FIG. 3) of operation **1004** or **1006** is replaced with one or more of layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A).

[0241] In some embodiments, operation **1006** is not performed. For example, in some embodiments, operation **1006** is not performed to manufacture integrated circuit **100** or **400**. In some embodiments, method **1000** is performed to manufacture an integrated circuit, **600B**, **700B** or **800B** based on one or more of layout designs **600A**, **700A** or **800A**, and operation **1006** is not performed on the manufactured integrated circuit.

[0242] FIGS. **11A-11B** is a flowchart of a method **1100** of generating a layout design of an integrated circuit in accordance with some embodiments. It is understood that additional operations may be performed before, during, and/or after the method **1100** depicted in FIGS. **11A-11B**, and that some other processes may only be briefly described herein. In some embodiments, the method

1100 is usable to generate one or more layout designs, such as layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A) of an integrated circuit, such as integrated circuit **100** (FIG. 1), integrated circuit **400** (FIGS. 4A-4E) or integrated circuit **500B**, **600B**, **700B**, **800B** or **900B**. In some embodiments, the method **1100** is usable to form integrated circuits having similar structural relationships as one or more of layout designs **200-300** or **500A-900A** (FIG. 2-3 or 5A-9A). In some embodiments, method **1100** is performed by a processing device (e.g., processor **1202** (FIG. 12)) configured to execute instructions for generating layout design **200** (FIG. 2), layout design **300** (FIG. 3), layout design **500A** (FIG. 5A), layout design **600A** (FIG. 6A), layout design **700A** (FIG. 7A), layout design **800A** (FIG. 8A) or layout design **900A** (FIG. 9A).

[0243] In operation **1102** of method **1100**, a set of active region layout patterns is generated. In some embodiments, the set of active region layout patterns of method **1100** includes one or more patterns of the set of active region layout patterns **202** of FIGS. 2-3 and 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0244] In operation **1104** of method **1100**, the set of active region layout patterns is placed on a first layout level of layout design **200**, **300** or **500A-900A**. In some embodiments, the first layout level corresponds to the OD layout level.

[0245] In operation **1106** of method **1100**, a set of fin layout patterns is generated. In some embodiments, the set of fin layout patterns of method **1100** includes one or more patterns of the set of fin layout patterns **316** of FIG. 3, and detailed description of these layout patterns is therefore omitted.

[0246] In operation **1108** of method **1100**, the set of fin layout patterns is placed on a second layout level of layout design **200**, **300** or **500A-900A**. In some embodiments, the second layout level corresponds to the FIN layout level. In some embodiments, the second layout level is the same as the first layout level. In some embodiments, the second layout level is different from the first layout level. In some embodiments, the second layout level is above the first layout level. In some embodiments, the second layout level is below the first layout level.

[0247] In operation **1110** of method **1100**, a first set of conductive feature layout patterns is generated. In some embodiments, the first set of conductive feature layout patterns of method **1100** includes one or more patterns of the set of conductive feature layout patterns **320**, **520**, **620**, **720**, **820** or **920** of FIGS. 3 & 5A-9A, and detailed description of these layout patterns is therefore omitted.

[0248] In operation **1112** of method **1100**, the first set of conductive feature layout patterns is placed on a first portion of the first layout level of layout design **200**, **300** or **500A-900A**. In some embodiments, the first portion of the first layout level corresponds to the MU layout level.

[0249] In operation **1114** of method **1100**, a set of gate layout patterns is generated. In some embodiments, the set of gate layout patterns of method **1100** includes one or more patterns of the set of gate layout patterns **205** of FIG. 2, set of gate layout patterns **206** of FIG. 2, set of gate layout patterns **204** of FIGS. 3 & 5A-7A, and set of gate layout patterns **804** of FIGS. 8A-9A, and detailed description of these layout patterns is therefore omitted.

[0250] In operation **1116** of method **1100**, the set of gate layout patterns is placed on a third layout level of layout design **200**, **300** or **500A-900A**. In some embodiments, the third layout level corresponds to the Poly layout level or MD layout level. In some embodiments, the third layout level is different from the first layout level and the second layout level. In some embodiments, the third layout level is above one or more of the first or second layout level. In some embodiments, the third layout level is below one or more of the first or second layout level.

[0251] In operation **1118** of method **1100**, a first set of via layout patterns is generated. In some embodiments, the first set of via layout patterns of method **1100** includes one or more patterns of the set of via layout patterns **322**, **522**, **622**, **822**, **922** of FIGS. 3 & 5A-9A, and detailed description

of these layout patterns is therefore omitted.

[0252] In operation **1120** of method **1100**, the first set of via layout patterns is placed between the first layout level and the third layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the first set of via layout patterns is placed on the VU layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the first set of via layout patterns is placed between the set of gate layout patterns and the first set of conductive feature layout patterns.

[0253] In operation **1122** of method **1100**, a set of metal over diffusion layout patterns is generated. In some embodiments, the set of metal over diffusion layout patterns of method **1100** includes one or more layout patterns of the set of metal over diffusion layout patterns **210, 610, 612, 810** or **910** of FIGS. **2-3** & **5A-9A**, and detailed description of these layout patterns is therefore omitted.

[0254] In operation **1124** of method **1100**, the set of metal over diffusion layout patterns is placed on the third layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the third layout level corresponds to the MD layout level.

[0255] In operation **1126** of method **1100**, a second set of via layout patterns is generated. In some embodiments, the second set of via layout patterns of method **1100** includes one or more patterns of the set of via layout patterns **322, 522, 622, 822, 922** of FIGS. **3** & **5A-9A**, and detailed description of these layout patterns is therefore omitted.

[0256] In operation **1128** of method **1100**, the second set of via layout patterns is placed between the third layout level and the first layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the second set of via layout patterns is placed on the VU layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the second set of via layout patterns is placed between the set of metal over diffusion layout patterns and the first set of conductive feature layout patterns.

[0257] In operation **1130** of method **1100**, a second set of conductive feature layout patterns is generated. In some embodiments, the second set of conductive feature layout patterns of method **1100** includes one or more patterns of the set of conductive feature layout patterns **330** of FIGS. **3** & **5A-9A**, or conductive feature layout pattern **630b, 630b'** of FIGS. **3** & **5A-9A**, and detailed description of these layout patterns is therefore omitted.

[0258] In operation **1132** of method **1100**, the second set of conductive feature layout patterns is placed on a fourth layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the fourth layout level is different from the first layout level, the second layout level and the third layout level. In some embodiments, the fourth layout level is above one or more of the first, second or third layout level. In some embodiments, the fourth layout level is below one or more of the first, second or third layout level. In some embodiments, the fourth layout level corresponds to the M0 layout level.

[0259] In operation **1134** of method **1100**, a third set of via layout patterns is generated. In some embodiments, the third set of via layout patterns of method **1100** includes one or more patterns of the set of via layout patterns **332, 632, 832** or via layout pattern **932a** of FIGS. **3** & **5A-9A**, and detailed description of these layout patterns is therefore omitted.

[0260] In operation **1136** of method **1100**, the third set of via layout patterns is placed between the third layout level and the fourth layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the third set of via layout patterns is placed on the VG layout level of layout design **200, 300** or **500A-900A**. In some embodiments, the third set of via layout patterns is placed between the set of gate layout patterns and the second set of conductive feature layout patterns.

[0261] In operation **1138** of method **1100**, a fourth set of via layout patterns is generated. In some embodiments, the fourth set of via layout patterns of method **1100** includes one or more patterns of the set of via layout patterns **636, 836** of FIGS. **6A-9A**, and detailed description of these layout patterns is therefore omitted.

[0262] In operation **1140** of method **1100**, the fourth set of via layout patterns is placed between the third layout level and the fourth layout level of layout design **200, 300** or **500A-900A**. In some

embodiments, the fourth set of via layout patterns is placed on the VD layout level of layout design **200**, **300** or **500A-900A**. In some embodiments, the fourth set of via layout patterns is placed between the set of metal over diffusion layout patterns and the second set of conductive feature layout patterns.

[0263] In operation **1142** of method **1100**, a third set of conductive feature layout patterns is generated. In some embodiments, the third set of conductive feature layout patterns of method **1100** includes one or more patterns of the set of conductive feature layout patterns **340**, **640** of FIGS. **3** & **5A-7A**, and detailed description of these layout patterns is therefore omitted.

[0264] In operation **1144** of method **1100**, the third set of conductive feature layout patterns is placed on a fifth layout level of layout design **200**, **300** or **500A-700A**. In some embodiments, the fifth layout level is different from the first layout level, the second layout level, the third layout level, and the fourth layout level. In some embodiments, the fifth layout level is above one or more of the first, second, third or fourth layout level. In some embodiments, the fifth layout level is below one or more of the first, second, third or fourth layout level. In some embodiments, the fifth layout level corresponds to the M1 layout level.

[0265] In operation **1146** of method **1100**, a fifth set of via layout patterns is generated. In some embodiments, the fifth set of via layout patterns of method **1100** includes one or more patterns of the set of via layout patterns **342**, **642** of FIGS. **3** & **5A-7A**, and detailed description of these layout patterns is therefore omitted.

[0266] In operation **1148** of method **1100**, the fifth set of via layout patterns is placed between the fourth layout level and the fifth layout level of layout design **200**, **300** or **500A-700A**. In some embodiments, the fifth set of via layout patterns is placed on the V0 layout level of layout design **200**, **300** or **500A-700A**. In some embodiments, the fifth set of via layout patterns is placed between the first set of conductive layout patterns and the second set of conductive feature layout patterns.

[0267] In operation **1150** of method **1100**, a cut feature layout pattern is generated. In some embodiments, the cut feature layout pattern of method **1100** includes cut feature layout pattern **304a** or **304b** (FIGS. **3** and **5A**) or cut feature layout pattern **904** (FIG. **9A**), and detailed description of these layout patterns is therefore omitted.

[0268] In operation **1152** of method **1100**, the cut feature layout pattern is placed on one or more of the third, fourth or fifth layout level of layout design **200**, **300** or **500A-900A**.

[0269] In some embodiments, one or more of the operations of method **1100** is not performed. One or more of the operations of methods **1000-1100** is performed by a processing device configured to execute instructions for manufacturing an integrated circuit, such as integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**. In some embodiments, one or more operations of methods **1000-1100** is performed using a same processing device as that used in a different one or more operations of methods **1000-1100**. In some embodiments, a different processing device is used to perform one or more operations of methods **1000-1100** from that used to perform a different one or more operations of methods **1000-1100**.

[0270] FIG. **12** is a schematic view of a system **1200** for designing an IC layout design in accordance with some embodiments. In some embodiments, system **1200** generates or places one or more IC layout designs described herein. System **1200** includes a hardware processor **1202** and a non-transitory, computer readable storage medium **1204** encoded, i.e., storing, with the computer program code **1206**, i.e., a set of executable instructions. Computer readable storage medium **1204** is configured for interfacing with manufacturing machines for producing the integrated circuit. The processor **1202** is electrically coupled to the computer readable storage medium **1204** via a bus **1208**. The processor **1202** is also electrically coupled to an I/O interface **1210** by bus **1208**. A network interface **1212** is also electrically connected to the processor **1202** via bus **1208**. Network interface **1212** is connected to a network **1214**, so that processor **1202** and computer readable storage medium **1204** are capable of connecting to external elements via network **1214**. The

processor **1202** is configured to execute the computer program code **1206** encoded in the computer readable storage medium **1204** in order to cause system **1200** to be usable for performing a portion or all of the operations as described in method **1000** or **1100**.

[0271] In some embodiments, the processor **1202** is a central processing unit (CPU), a multi-processor, a distributed processing system, an application specific integrated circuit (ASIC), and/or a suitable processing unit.

[0272] In some embodiments, the computer readable storage medium **1204** is an electronic, magnetic, optical, electromagnetic, infrared, and/or a semiconductor system (or apparatus or device). For example, the computer readable storage medium **1204** includes a semiconductor or solid-state memory, a magnetic tape, a removable computer diskette, a random access memory (RAM), a read-only memory (ROM), a rigid magnetic disk, and/or an optical disk. In some embodiments using optical disks, the computer readable storage medium **1204** includes a compact disk-read only memory (CD-ROM), a compact disk-read/write (CD-R/W), and/or a digital video disc (DVD).

[0273] In some embodiments, the storage medium **1204** stores the computer program code **1206** configured to cause system **1200** to perform method **1000** or **1100**. In some embodiments, the storage medium **1204** also stores information needed for performing method **1000** or **1100** as well as information generated during performing method **1000** or **1100**, such as layout design **1216** and user interface **1218**, and/or a set of executable instructions to perform the operation of method **1000** or **1100**. In some embodiments, layout design **1216** comprises one or more of layout designs **200-300** or **500A-900A**. In some embodiments, layout design **1216** comprises one or more layout patterns of layout designs **200-300** or **500A-900A**.

[0274] In some embodiments, the storage medium **1204** stores instructions (e.g., computer program code **1206**) for interfacing with manufacturing machines. The instructions (e.g., computer program code **1206**) enable processor **1202** to generate manufacturing instructions readable by the manufacturing machines to effectively implement method **1000** or **1100** during a manufacturing process.

[0275] System **1200** includes I/O interface **1210**. I/O interface **1210** is coupled to external circuitry. In some embodiments, I/O interface **1210** includes a keyboard, keypad, mouse, trackball, trackpad, and/or cursor direction keys for communicating information and commands to processor **1202**.

[0276] System **1200** also includes network interface **1212** coupled to the processor **1202**. Network interface **1212** allows system **1200** to communicate with network **1214**, to which one or more other computer systems are connected. Network interface **1212** includes wireless network interfaces such as BLUETOOTH, WIFI, WIMAX, GPRS, or WCDMA; or wired network interface such as ETHERNET, USB, or IEEE-1394. In some embodiments, method **1000** or **1100** is implemented in two or more systems **1200**, and information such as layout design, and user interface are exchanged between different systems **1200** by network **1214**.

[0277] System **1200** is configured to receive information related to a layout design through I/O interface **1210** or network interface **1212**. The information is transferred to processor **1202** by bus **1208** to determine a layout design for producing an integrated circuit, e.g., integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**. The layout design is then stored in computer readable medium **1204** as layout design **1216**. System **1200** is configured to receive information related to a user interface through I/O interface **1210** or network interface **1212**. The information is stored in computer readable medium **1204** as user interface **1218**.

[0278] In some embodiments, method **1000** or **1100** is implemented as a standalone software application for execution by a processor. In some embodiments, method **1000** or **1100** is implemented as a software application that is a part of an additional software application. In some embodiments, method **1000** or **1100** is implemented as a plug-in to a software application. In some embodiments, method **1000** or **1100** is implemented as a software application that is a portion of an EDA tool. In some embodiments, method **1000** or **1100** is implemented as a software application

that is used by an EDA tool. In some embodiments, the EDA tool is used to generate a layout of the integrated circuit device. In some embodiments, the layout is stored on a non-transitory computer readable medium. In some embodiments, the layout is generated using a tool such as VIRTUOSO® available from CADENCE DESIGN SYSTEMS, Inc., or another suitable layout generating tool. In some embodiments, the layout is generated based on a netlist which is created based on the schematic design. In some embodiments, method **1000** or **1100** is implemented by a manufacturing device to manufacture an integrated circuit (e.g., integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**) using a set of masks manufactured based on one or more layout designs (e.g., layout design **200-300** or **500A-900A**) generated by system **1200**. System **1200** of FIG. **12** generates layout designs (e.g., layout design **200-300** or **500A-900A**) of an integrated circuit (e.g., integrated circuit **100**, **400**, **500B**, **600B**, **700B**, **800B** or **900B**) that are smaller than other approaches.

[0279] One aspect of this description relates to an integrated circuit. In some embodiments, the integrated circuit includes a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction. In some embodiments, the integrated circuit further includes a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region. In some embodiments, the integrated circuit further includes a first contact extending in the second direction, overlapping the first conductor and being located on a second level different from the first level. In some embodiments, the integrated circuit further includes a second contact extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first contact in at least the first direction. In some embodiments, the integrated circuit further includes a set of gates extending in the second direction, overlapping the first conductor, and being located on the second level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction. In some embodiments, the first conductor electrically couples the first contact and the second contact together.

[0280] Another aspect of this description relates to an integrated circuit. In some embodiments, the integrated circuit includes a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction. In some embodiments, the integrated circuit further includes a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region. In some embodiments, the integrated circuit further includes a first gate extending in the second direction, overlapping the first conductor and being located on a second level different from the first level. In some embodiments, the integrated circuit further includes a second gate extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first gate in at least the first direction. In some embodiments, the integrated circuit further includes a set of contacts extending in the second direction, overlapping the first conductor, and being located on the second level, each of the contacts of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction. In some embodiments, the first conductor electrically couples the first gate and the second gate together.

[0281] Still another aspect of this description relates to an integrated circuit. In some embodiments, the integrated circuit includes a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction. In some embodiments, the integrated circuit further includes a first conductive structure extending in the first direction, being located on the first level, and being between the first active region and

the second active region. In some embodiments, the integrated circuit further includes a set of gates extending in the second direction, overlapping at least the first conductive structure and being located on a second level different from the first level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction by a first pitch. In some embodiments, the integrated circuit further includes a set of contacts extending in the second direction, overlapping the first conductive structure, and being located on the second level, each contact of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction, and a first contact of the set of contacts being directly coupled to a first source/drain of the first active region. In some embodiments, the integrated circuit further includes a first set of vias coupling the first conductive structure to the set of contacts, the first set of vias being between the first conductive structure and the set of contacts, and a first via of the first set of vias being located where the first contact of the set of contacts overlaps the first conductive structure.

[0282] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. An integrated circuit, comprising: a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction; a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region; a first contact extending in the second direction, overlapping the first conductor and being located on a second level different from the first level; a second contact extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first contact in at least the first direction; and a set of gates extending in the second direction, overlapping the first conductor, and being located on the second level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction, wherein the first conductor electrically couples the first contact and the second contact together.
2. The integrated circuit of claim 1, wherein the first conductor comprises: a first portion extending in the first direction, being located on the lower portion of the first level, and being electrically coupled to the first contact; a second portion extending in the first direction, being located on the lower portion of the first level, being electrically coupled to the second contact, and being separated from the first portion in the second direction; and a third portion extending in the second direction, being located on the lower portion of the first level, being coupled to the first portion and the second portion, and being between the first portion and the second portion.
3. The integrated circuit of claim 2, further comprising: a first via coupling the first portion to the first contact, and being between the first portion and the first contact, wherein the first contact overlaps the first portion.
4. The integrated circuit of claim 3, further comprising: a second via coupling the second portion to the second contact, and being between the second portion and the second contact, wherein the second contact overlaps the second portion.
5. The integrated circuit of claim 4, further comprising: a third contact extending in the second direction, being separated from the first contact in the second direction, being separated from the

second contact in the first direction, and being located on the second level; and a fourth contact extending in the second direction, being separated from the first contact in the first direction, being separated from the second contact in the second direction, and being located on the second level.

6. The integrated circuit of claim 5, further comprising: a second conductor extending in the first direction, being between the first active region and the second active region, overlapping the first contact and the fourth contact, and being located on a third level different from the first level and the second level; a third conductor extending in the first direction, being between the first active region and the second active region, overlapping the second contact and the third contact, being located on the third level, and being separated from the second conductor in the second direction; and a fourth conductor extending in the first direction, being between the third conductor and the fourth conductor, overlapping each of the first conductor, a first gate of the set of gates and a second gate of the set of gates, being located on the third level, and being separated from the second conductor and the third conductor in the second direction.

7. The integrated circuit of claim 6, further comprising: a third via coupling the second conductor to the fourth contact, and being between the second conductor and the fourth contact; and a fourth via coupling the third conductor to the third contact, and being between the third conductor and the third contact.

8. The integrated circuit of claim 7, further comprising: a fifth via coupling the fourth conductor to the first gate of the set of gates, and being between the fourth conductor and the first gate of the set of gates; and a sixth via coupling the fourth conductor to the second gate of the set of gates, and being between the fourth conductor and the second gate of the set of gates.

9. An integrated circuit, comprising: a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction; a first conductor extending in the first direction and the second direction, being located on a lower portion of the first level, and being between the first active region and the second active region; a first gate extending in the second direction, overlapping the first conductor and being located on a second level different from the first level; a second gate extending in the second direction, overlapping the first conductor, being located on the second level, and being separated from the first gate in at least the first direction; and a set of contacts extending in the second direction, overlapping the first conductor, and being located on the second level, each of the contacts of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction, wherein the first conductor electrically couples the first gate and the second gate together.

10. The integrated circuit of claim 9, wherein the first conductor comprises: a first portion extending in the first direction, being located on the lower portion of the first level, and being electrically coupled to the first gate; a second portion extending in the first direction, being located on the lower portion of the first level, being electrically coupled to the second gate, and being separated from the first portion in the second direction; and a third portion extending in the second direction, being located on the lower portion of the first level, being coupled to the first portion and the second portion, and being between the first portion and the second portion.

11. The integrated circuit of claim 10, further comprising: a first via coupling the first portion to the first gate, and being between the first portion and the first gate, wherein the first gate overlaps the first portion; and a second via coupling the second portion to the second gate, and being between the second portion and the second gate, wherein the second gate overlaps the second portion.

12. The integrated circuit of claim 11, further comprising: a third gate extending in the second direction, overlapping the second active region, being separated from the first gate in the second direction, being separated from the second gate in the first direction, and being located on the second level; and a fourth gate extending in the second direction, overlapping the first active region, being separated from the first gate in the first direction, being separated from the second

gate in the second direction, and being located on the second level.

13. The integrated circuit of claim 12, further comprising: a second conductor extending in the first direction, being between the first active region and the second active region, overlapping the first gate and the fourth gate, and being located on a third level different from the first level and the second level; a third conductor extending in the first direction, being between the first active region and the second active region, overlapping the second gate and the third gate, being located on the third level, and being separated from the third conductor in the second direction; and a fourth conductor extending in the first direction, being between the second conductor and the third conductor, overlapping the first conductor, being located on the third level, and being separated from the second conductor and the third conductor in the second direction.

14. The integrated circuit of claim 13, further comprising: a third via coupling the third conductor to the third gate, and being between the third conductor and the third gate; and a fourth via coupling the second conductor to the fourth gate, and being between the second conductor and the fourth gate.

15. An integrated circuit, comprising: a first active region and a second active region in a substrate, the first active region and the second active region extending in a first direction, being located on a first level, and being separated from one another in a second direction different from the first direction; a first conductive structure extending in the first direction, being located on the first level, and being between the first active region and the second active region; a set of gates extending in the second direction, overlapping at least the first conductive structure and being located on a second level different from the first level, each gate of the set of gates being separated from an adjacent gate of the set of gates in the first direction by a first pitch; a set of contacts extending in the second direction, overlapping the first conductive structure, and being located on the second level, each contact of the set of contacts being separated from an adjacent contact of the set of contacts in the first direction, and a first contact of the set of contacts being directly coupled to a first source/drain of the first active region; and a first set of vias coupling the first conductive structure to the set of contacts, the first set of vias being between the first conductive structure and the set of contacts, and a first via of the first set of vias being located where the first contact of the set of contacts overlaps the first conductive structure.

16. The integrated circuit of claim 15, further comprising: a second conductive structure extending in the first direction, being located on the first level, being between the first active region and the second active region, and being separated from the first conductive structure in the second direction.

17. The integrated circuit of claim 16, further comprising: a second set of vias coupling the second conductive structure to the set of contacts, the second set of vias being between the second conductive structure and the set of contacts, and a first via of the second set of vias being located where a second contact of the set of contacts overlaps the second conductive structure.

18. The integrated circuit of claim 17, further comprising: a first set of conductive structures extending in the first direction, being between the first active region and the second active region, overlapping at least the set of gates, and being located on a third level different from the first level and the second level, and each conductive structure of the first set of conductive structures being separated from an adjacent conductive structure of the first set of conductive structures in the second direction.

19. The integrated circuit of claim 18, further comprising: a second set of vias coupling the first set of conductive structures to the set of contacts, the second set of vias being between the first set of conductive structures and the set of contacts, and a first via of the second set of vias being located where a first conductive structure of the first set of conductive structures overlaps a second contact of the set of contacts; and a third set of vias coupling the first set of conductive structures to the set of gates, the third set of vias being between the first set of conductive structures and the set of gates, and a first via of the third set of vias being located where a second conductive structure of

the first set of conductive structures overlaps a first gate of the set of gates.

20. The integrated circuit of claim 19, further comprising: a second conductive structure extending in the second direction, being between the set of gates, overlapping at least the first conductive structure, and being located on a fourth level different from the first level, the second level and the third level; and a fourth set of vias coupling the second conductive structure to the first set of conductive structures, the fourth set of vias being between the second conductive structure and the first set of conductive structures, and a first via of the fourth set of vias being located where the second conductive structure overlaps the first conductive structure of the first set of conductive structures.
